

TECHNICAL SPECIFICATION



**Safety of machinery – Electro-sensitive protective equipment –
Part 4-3: Particular requirements for equipment using vision based protective
devices (VBPD) – Additional requirements when using stereo vision techniques
(VBPDST)**



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Part 4-3: Particular requirements for equipment using vision based protective
devices (VBPD) – Additional requirements when using stereo vision techniques
(VBPDST)**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**SAFETY OF MACHINERY –
ELECTRO-SENSITIVE PROTECTIVE EQUIPMENT –****Part 4-3: Particular requirements for equipment using
vision based protective devices (VBPD) –
Additional requirements when using stereo
vision techniques (VBPDST)**

FOREWORD

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- the subject is still under technical development or where, for any other reason, there is the future but no immediate possibility of an agreement on an International Standard.

Technical specifications are subject to review within three years of publication to decide whether they can be transformed into International Standards.

IEC TS 61496-4-3, which is a technical specification, has been prepared by IEC technical committee 44: Safety of machinery – Electrotechnical aspects.

The text of this technical specification is based on the following documents:

Enquiry draft	Report on voting
44/711/DTS	44/722/RVC

Full information on the voting for the approval of this technical specification can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

This part is to be used in conjunction with IEC 61496-1:2012.

This part supplements or modifies the corresponding clauses in IEC 61496-1:2012 to specify particular requirements for the design, construction and testing of electro-sensitive protective equipment (ESPE) for the safeguarding of machinery, employing vision based protective devices (VBPD) using stereo vision techniques (VBPDST) for the sensing function.

Where a particular clause or subclause of Part 1 is not mentioned in this Part 4-3, that clause or subclause applies as far as is reasonable. Where this part states "*addition*", "*modification*" or "*replacement*", the relevant text of Part 1 shall be adapted accordingly.

Clauses and subclauses which are additional to those of Part 1 are numbered sequentially, following on the last available number in Part 1. Terminological entries (in Clause 3) which are additional to those in Part 1 are numbered starting from 3.4301. Additional annexes are lettered from AA onwards.

A list of all parts in the IEC 61496 series, published under the general title *Safety of machinery – Electro-sensitive protective equipment*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

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- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

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INTRODUCTION

An electro-sensitive protective equipment (ESPE) is applied to machinery presenting a risk of personal injury. It provides protection by causing the machine to revert to a safe condition before a person can be placed in a hazardous situation.

The working group responsible for drafting this technical specification was concerned that, due to the complexity of the technology, there are many issues that are highly dependent on analysis and expertise in specific test and measurement techniques. In order to provide a high level of confidence, independent review by relevant expertise is required. They considered that if this high level of confidence could not be established these devices would not be suitable for use in safety related applications.

SAFETY OF MACHINERY – ELECTRO-SENSITIVE PROTECTIVE EQUIPMENT –

Part 4-3: Particular requirements for equipment using vision based protective devices (VBPD) – Additional requirements when using stereo vision techniques (VBPDST)

1 Scope

Replacement:

This part of IEC 61496 specifies requirements for the design, construction and testing of electro-sensitive protective equipment (ESPE) designed specifically to detect persons or parts of persons as part of a safety-related system, employing vision-based protective devices (VBPDs) using stereo vision techniques (VBPDST) for the sensing function. Special attention is directed to features which ensure that an appropriate safety-related performance is achieved. An ESPE may include optional safety-related functions, the requirements for which are given in Annex A of IEC 61496-1:2012 and this Technical Specification.

This part of IEC 61496 does not specify the dimensions or configurations of the detection zone and its disposition in relation to hazardous parts for any particular application, nor what constitutes a hazardous state of any machine. It is restricted to the functioning of the ESPE and how it interfaces with the machine.

The detection principle is based on the evaluation of images from different viewing points (stereoscopic view) for the determination of distance information. This distance information is used to determine the location of an object(s).

- This part of IEC 61496 is limited to vision based ESPEs with distances (stereo base) and directions between the different imaging devices fixed during manufacture.
- It is limited to vision based ESPEs, with a minimum distance from the sensing device to the detection zone of 4 times of the stereo base.
- It is limited to vision based ESPEs that can detect objects with at least 5 pixel diameter in the image plane.
- It is limited to vision based ESPEs that do not require human intervention for detection.
- It is limited to vision based ESPEs that detect objects entering into or being present in a detection zone(s).
- It is limited to VBPDSTs employing radiation at wavelengths within the range 400 nm to 1 500 nm.
- This part of IEC 61496 does not address those aspects required for complex classification or differentiation of the object detected.
- This part of IEC 61496 does not consider the aspects of a moving ESPE installation.

Additional requirements and tests can apply in the following cases:

- Use of multi-spectral (colour) techniques;
- Setups other than as shown in Figures of 4.1.2 (e.g. changing backgrounds, horizontal orientation of the optical axis with respect to the floor);
- Intended for outdoor applications.

This technical specification is relevant for VBPDSTs having a stated detection capability up to 200 mm.

This technical specification may be relevant to applications other than those for the protection of persons or parts of persons like arm or fingers (in the range 14 mm to 200 mm), for example the protection of machinery or products from mechanical damage. In those applications, additional requirements can be necessary, for example when the materials that are to be recognized by the sensing function have different properties from those of persons.

This technical specification does not deal with EMC emission requirements.

2 Normative references

Addition:

IEC 60825-1:2014, *Safety of laser products – Part 1 – Equipment classification and requirements*

IEC 61496-1:2012, *Safety of machinery – Electro-sensitive protective equipment – Part 1: General requirements and tests*

IEC 62471, *Photobiological safety of lamps and lamp systems*

ISO 13855:2010, *Safety of machinery – Positioning of safeguards with respect to the approach speeds of parts of the human body*

ISO 20471, *High visibility clothing – Test methods and requirements*

3 Terms and definitions

Replacement:

3.3

detection capability

ability to detect the specified test pieces (see 4.2.13) in the specified detection zone

Note 1 to entry: Detection capability is measured by the size of an object that can be detected. An increase in detection capability means that a smaller object can be detected.

[SOURCE: IEC 61496-1:2012, 3.3, modified – text changed to make more relevant to vision based sensors.]

3.4

detection zone, <of a VBPDST>

three-dimensional volume (for example, in the shape of a pyramid or cone) within which a specified test piece will be detected by the VBPDST

3.5

electro-sensitive protective equipment ESPE

Addition:

Note 3 to entry: Illumination unit(s), if applicable, is/are part(s) of the sensing device.

Additional definitions:

3.4301

minimum detection zone

lowest dimension of the detection zone for a test piece moving with maximum speed

Note 1 to entry: This is the lowest dimension that ensures the integrity of the detection capability.

3.4302

evaluation images, pl

set of images which are used by the detection algorithms

SEE: Figure 1.

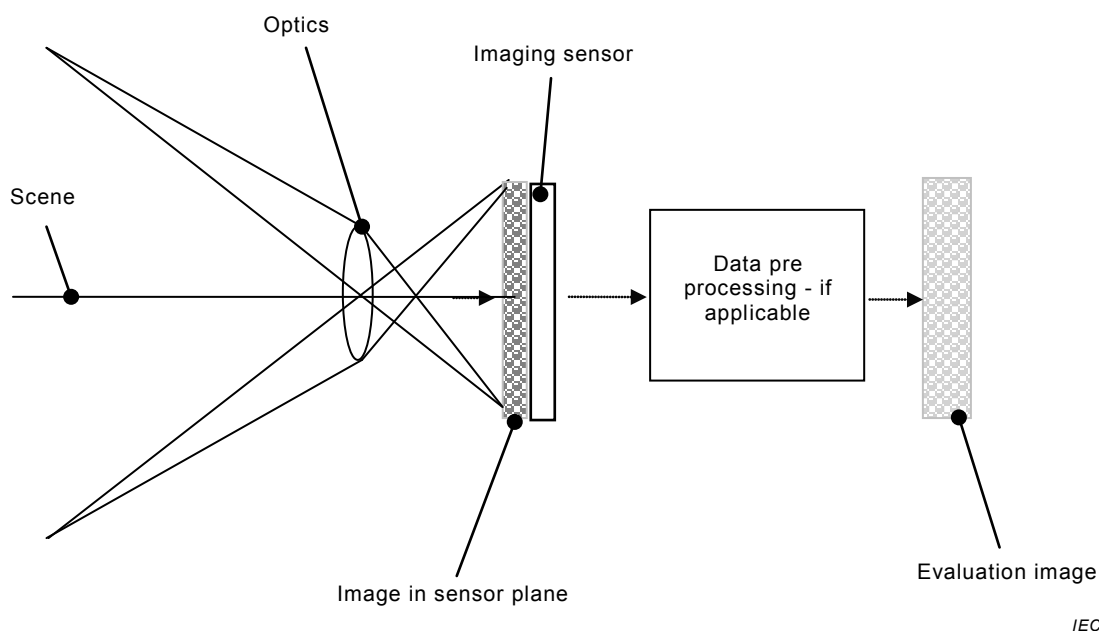


Figure 1 – Image planes in imaging device of a VBPDST

3.4303

image, <of a VBPDST>

snapshot representation of the scene in different planes of the VBPDST in form of a two dimensional pixel matrix

3.4304

vision based protective device using stereo vision techniques

VBPDST

VBPD with two or more imaging devices using stereo vision techniques

3.4305

imaging sensor

opto-electronic device which produces electrical signals representing the characteristics of an image

SEE: Figure 1.

3.4306

imaging device

combination of an imaging sensor, optics and the processing unit (if applicable)

SEE: Figure 1.

Note 1 to entry: The imaging devices are part of the sensing device.

3.4307**operating distance**

distance measured along the z-axis of the sensing device coordinate system

3.4308**pixel, <of a sensor>**

smallest light sensitive element of an imaging sensor

3.4309**pixel, <of an image>**

area of the smallest element that can be distinguished from its neighbouring elements

3.4310**ambient illumination technique****AIT**

technique that relies on scene lighting for illumination and contrast to obtain range measurements

3.4311**pattern projection technique****PAPT**

technique that uses a special projection to enhance the contrast of a scene

3.4312**sensing device coordinate system**

coordinate system oriented to the sensing device

Note 1 to entry: Typically the z-axis is parallel to the optical axis of one imaging device.

3.4313**tolerance zone**

zone outside of and adjacent to the detection zone within which the specified test piece is detected with a probability of detection lower than the required probability within the detection zone

Note 1 to entry: The tolerance zone is necessary to achieve the required probability of detection of the specified test piece within the detection zone. For explanation of the concept of probability of detection and the tolerance zone, see Annex BB.

3.4314**user coordinate system**

coordinate system that may be configured by the user

3.4315**zone with limited detection capability**

volume between the detection zone and the front of the sensing device in which the stated detection capability is not achieved

3.4316**stereo base**

distance between the entrance pupils of two imaging devices

Note 1 to entry: The expression baseline is often used as synonym for stereo base.

3.4317**position accuracy**

accuracy in three dimensions of the position of an object as measured by VBPDST

Addition:

Abbreviated terms

AIT	Ambient illumination technique
BTP	Black test piece
GB	Grey background
GTP	Grey test piece
lx	Lux
LC	Low contrast
OD	Operating distance
P1	Position 1 of the light source
P2	Position 2 of the light source
PAPT	Pattern projection techniques
PTZ	Tolerance zone related to probability
RRTP	Retro-reflective test piece
STZ	Tolerance zone related to systematic influences
TTC	Typical test condition (test condition for normal operation tests)
TI	Typical illumination (illumination used for normal operation tests)
VBPDEST	Vision based protective devices using stereo vision techniques
WTP	White test piece

4 Functional, design and environmental requirements

This clause of Part 1 is applicable except as follows:

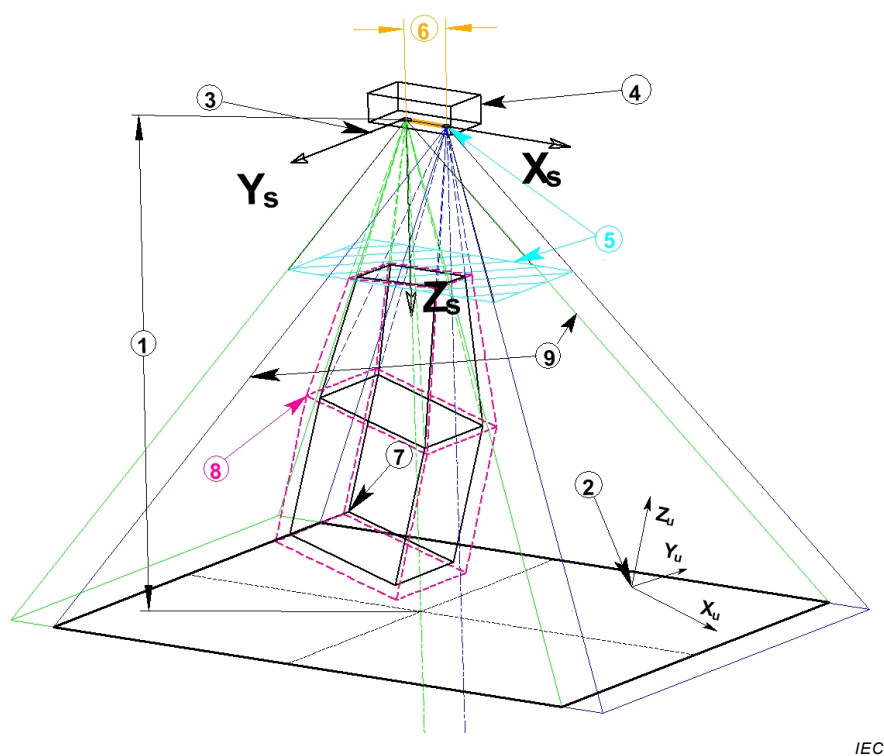
4.1 Functional requirements

Replacement:

4.1.2 Sensing function

The detection zone shall begin at the border of the zone with limited detection capability and end at the maximum operating distance (see Figure 2 and Figure 3).

Object(s) in the zone with limited detection capability shall not reduce the detection capability within the detection zone. Any reduction of the detection capability shall be detected and the VBPDEST shall go to lock-out condition (see 4.2.2.4).



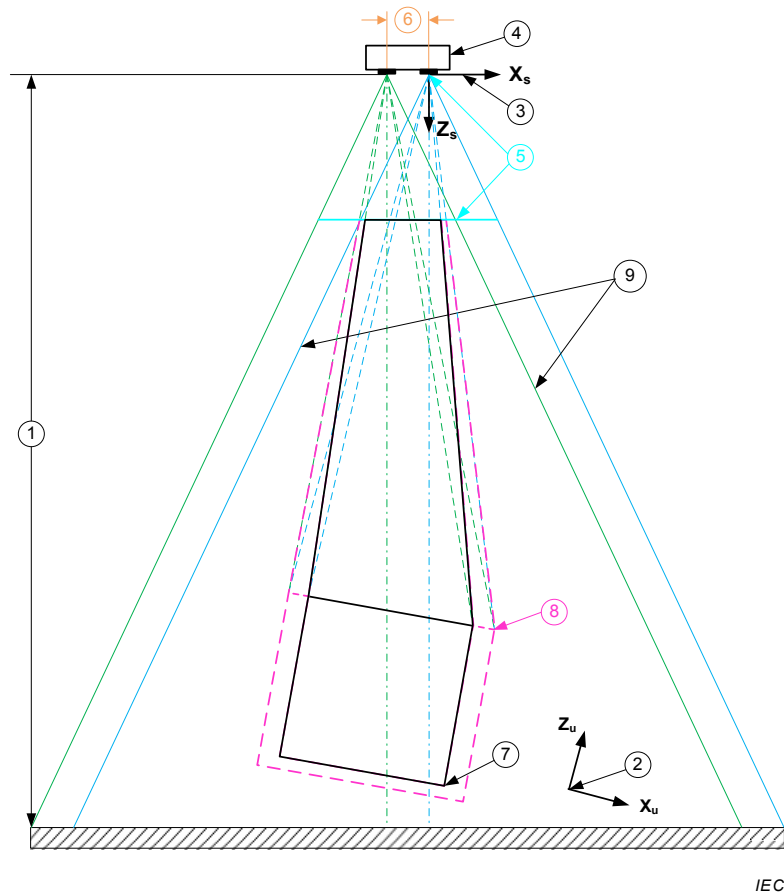
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Key

- | | | |
|--------------------------------------|--|--------------------------|
| 1 - Maximum operating distance | 4 - Sensing device | 7 - Detection zone |
| 2 - User coordinate system | 5 - Zone with limited detection capability | 8 - Tolerance zone |
| 3 - Sensing device coordinate system | 6 - Stereo base | 9 - Stereo field of view |

NOTE The figure shows a system with parallel axes and a maximum operating distance on a plane perpendicular to the axes.

Figure 2 – 3D view of a vision based protective device using stereo vision techniques (VBPDS)



Key

- | | | |
|-------------------------------------|---|-------------------------|
| 1- Maximum operating distance | 4- Sensing device | 7- Detection zone |
| 2- User coordinate system | 5- Zone with limited detection capability | 8- Tolerance zone |
| 3- Sensing device coordinate system | 6- Stereo base | 9- Stereo field of view |

NOTE This figure shows a system with parallel axes and a maximum operating distance on a plane perpendicular to the axes.

Figure 3 – 2D view of a vision based protective device using stereo vision techniques (VBPDS)

*Additional functional requirements:***4.1.2.1 General**

The sensing function shall be effective over the detection zone. No adjustment of the detection zone or detection capability shall be possible without the use of a security measure (e.g. key, keyword, or tool).

The VBPDST shall respond by giving appropriate output signal(s) when a test piece is present anywhere within the detection zone whether static or moving with respect to the VBPDST.

The supplier shall specify the limits of detection capability. The supplier shall take into account worst case scenario considering all influences listed in this technical specification including, for example:

- signal-to-noise ratio;
- light intensity in the image in sensor plane (see Figure 1);
- contrast on the image in sensor plane;
- position of the image in sensor plane;

4.1.2.2 Optical performance

The VBPDST shall be designed and constructed to:

- a) limit the possibility of malfunction during exposure to extraneous radiation in the range of 400 nm to 1 500 nm;
- b) limit the effects of environmental influences (temperature, vibration and bumps, dust, moisture, ambient light, extraneous reflections, changing illumination, shadows on background, background reflectivity);
- c) limit the misalignment at which normal operation is possible.

4.1.3 Types of ESPE*Replacement:*

In this part of IEC 61496, only a type 3 ESPE is considered. It is the responsibility of the machine supplier and/or the user to prescribe if this type is suitable for a particular application.

The type 3 ESPE shall fulfil the fault detection requirements of 4.2.2.4 of this part of IEC 61496. In normal operation, the output circuit of each of at least two output signal switching devices (OSSDs) of the type 3 ESPE shall go to the OFF-state when the sensing device is actuated, or when the power is removed from the device.

*Addition:***4.1.6 Zone with limited detection capability**

A zone between the optical window and the beginning of the detection zone is referred to as a zone with limited detection capability. In order to ensure that no hazard can arise in a particular application due to the presence of objects in the zone between the optical window and the detection zone, its dimensions and appropriate information for use shall be provided by the supplier.

4.2 Design requirements

4.2.2 Fault detection requirements

4.2.2.2 Particular requirements for a type 1 ESPE

This subclause of Part 1 is not applicable.

4.2.2.3 Particular requirements for a type 2 ESPE

This subclause of Part 1 is not applicable.

4.2.2.4 Particular requirements for a type 3 ESPE

Replacement:

A single fault in the sensing device resulting in a complete loss of the stated VBPDEST detection capability shall cause the ESPE to go to a lock-out condition within the specified response time.

A single fault resulting in a deterioration of the stated VBPDEST detection capability shall cause the ESPE to go to the lock-out condition within a time period of 5 s following the occurrence of that fault.

NOTE Examples of deterioration of the VBPDEST detection capability include:

- increase of the minimum detectable object size;
- increase in the minimum detectable contrast;
- decrease of position accuracy.

A single fault resulting in an increase in response time beyond the specified value or preventing at least one OSSD going to the OFF-state shall cause the ESPE to go to a lock-out condition immediately, i.e. within the response time, or immediately upon any of the following demand events where fault detection requires a change in state:

- on actuation of the sensing function;
- on switch off/on;
- on reset of the start interlock or the restart interlock, if provided (see Clauses A.5 and A.6 of IEC 61496-1:2012).

It shall not be possible for the ESPE to achieve a reset from a lock-out condition, for example, by interruption and restoration of the mains power supply or by any other means, when the fault which initiated the lock-out condition is still present.

In cases where a single fault which does not cause a failure to danger of the ESPE is not detected, the occurrence of one additional fault shall not cause a failure to danger.

For verification of this requirement, see 5.3.4.

4.2.2.5 Particular requirements for a type 4 ESPE

This subclause of Part 1 is not applicable.

Additional design requirements:

4.2.12 Integrity of the VBPDEST detection capability

4.2.12.1 General

The design of the VBPDEST shall ensure that the detection capability is not degraded below the limits specified by the supplier and in this Technical Specification by any of, but not limited to, the following:

- a) low contrast between an object and background on the evaluation images;
- b) the position of the object within the detection zone;
- c) the number of objects within the detection zone;
- d) the size of object(s) within the detection zone;
- e) auto-adjustment of optical and electrical characteristics;
- f) properties/limitations of optical and electrical components;
- g) accuracy of object position in image(s);
- h) at the limits of alignment and/or adjustment;
- i) ageing of components;
- j) performance and limitations of the optical components;
- k) component tolerances;
- l) changing of internal and external references to ensure the detection capability;
- m) environmental conditions specified in 4.3.

If a single fault, which under normal operating conditions (see 5.1.2.1) would not result in a loss of the stated VBPDEST detection capability but, when occurring with a combination of the above conditions, would result in such a loss, that fault together with that combination of conditions (as determined to be relevant during the analysis of the design) shall be considered as a single fault and the VBPDEST shall respond to such a single fault as required in 4.2.2.4.

4.2.12.2 Object detection at low contrast

At low contrast the test piece shall be detected when the VBPDEST is in normal operation.

NOTE A physical contrast results in a difference of intensity that is detected. For more information, see Annex CC.

4.2.12.3 Object detection at high contrast

At high contrast the test piece shall be detected when the VBPDEST is in normal operation.

NOTE A high contrast results from a big difference of the diffuse reflectance value and/or lighting variation between the background and the test piece. The contrast could be higher than the dynamic range of the imaging sensor.

4.2.12.4 Minimum detection zone

The supplier shall specify the minimum detection zone(s). The supplier shall take into account worst case conditions including, for example:

- response time;
- minimum diameter of the test piece;
- maximum speed of the test piece.

4.2.12.5 Response time

Objects of the minimum detectable size that are either stationary or moving within the detection zone at any speed up to 1,6 m/s shall be detected by the ESPE within the specified response time. The supplier shall specify the maximum response time. The supplier shall take

into account worst case conditions including, for example, frame rate, evaluation time, minimum diameter of the test piece, maximum speed of the test piece, number of objects in the detection zone and the values of the minimum detection zone, as well as environmental influences. Where the supplier states that a VBPDST can be used to detect objects moving at speeds greater than 1,6 m/s, the requirements shall be met at any speed up to and including the stated maximum speed.

4.2.12.6 Detection zone(s) and tolerance zone(s)

The supplier shall define values up to 200 mm as the minimum detectable object size of the VBPDST within the detection zone. The minimum detectable object size may be distance dependent.

The test pieces (see 4.2.13) shall be detected with a minimum probability of detection of $1 - 2,9 \times 10^{-7}$ throughout the detection zone(s). To achieve this minimum probability of detection, the tolerance zone has to be considered in addition to the detection zone. The tolerance zone depends on position accuracy composed of systematic (STZ) and random influences (PTZ). Even if a measured distance value of a test piece falls into the tolerance zone, this test piece will be determined as detected and the OSSDs shall go to the OFF-state or remain in the OFF-state. If a part of the position error does not result in failure to danger, then it does not need to be included in the tolerance zone.

NOTE 1 Under the assumption that errors are normally distributed, the PTZ will be 5 times the standard deviation of the error distribution (5 sigma).

NOTE 2 Within the tolerance zone there is no requirement for maintaining a minimum probability of detection.

NOTE 3 The dimensions of the tolerance zone in the three axes can be different.

NOTE 4 An example to determine the tolerance zone is given in Annex BB.

The supplier shall specify the tolerance zone(s) and take into account worst-case considerations according to the influences listed in 4.2.12.7.

When using reference markers or parts of the environment such as walls or the floor during setup, errors in determining the correct distance and position of these references shall be taken into account when specifying the tolerance zone(s).

NOTE 5 The dimension of the tolerance zone can be influenced by the method of approach (for example walking, crawling, sliding along a wall). If such information is used to calculate the tolerance zone, then appropriate analysis or tests can be required.

NOTE 6 If the required probability of detection can be satisfied by partial intrusion of an object into the detection zone, then a smaller tolerance zone can be used for safety distance determination. Further details can be found in Annex AA.

4.2.12.7 Position accuracy

When determining the position accuracy, the following systematic and random influences shall be taken into account:

- a) the calibration of the sensing device;
- b) characteristics of the optical/imaging sensor, such as
 - 1) the number of pixels and pixel size
 - 2) signal-to-noise ratio
 - 3) modulation transfer function of the optics
- c) algorithmic influences, such as:
 - 1) smoothing algorithm
 - 2) feature based detection algorithm, e.g. edge detection algorithm
 - 3) template matching

- 4) colour sensor and algorithm
- 5) global algorithms, e.g. cluster algorithm
- 6) optical flow analysis algorithm
- 7) object tracking algorithm
- 8) stereo algorithm
- d) synchronization between imaging sensors;
- e) characteristics of the test piece;
- f) limits of illumination;
- g) Ageing and tolerances of components and references.

4.2.12.8 Pattern projection techniques (PAPT)

If pattern projection techniques are used to enhance the contrast in the scene, then the pattern projector shall be considered as part of the VBPDST. This illumination module shall have the capability to project sufficient contrast onto the scene anywhere within the specified detection zone such that the system fulfils the requirements of this Technical Specification. The detection capability shall not be decreased below the limits specified by the supplier by any, but not limited to the following influences:

- a) contrast between projected pattern elements;
- b) contrast changes within projected pattern elements;
- c) size and differentiation of projected pattern elements and number of projected pattern elements used for contrast enhancement;
- d) size of pixels and numbers of pixels compared to used projected pattern elements;
- e) automatic adaptation of algorithm/routines;
- f) size, shape, colour, reflectivity, position and surface structure of object and scene compared to projected pattern;
- g) resulting superposition of natural object/scene contrast and contrast produced by pattern projection;
- h) position and location of pattern projector.

4.2.12.9 Influence of periodic surface structures on the background

Periodic surface structures on the background shall not lead to a failure to danger.

Periodic surface structures resulting in a complete loss of the stated VBPDST detection capability shall cause the ESPE to go to a lock-out condition within the specified response time.

Periodic surface structures resulting in a deterioration of the stated VBPDST detection capability shall cause the ESPE to go to the lock-out condition within a time period of 5 s following the occurrence of periodic surface structures.

These requirements are verified by the tests of 5.2.1.5.

4.2.13 Test pieces for type testing

4.2.13.1 General

The test pieces shall be provided by the supplier for use in the type tests of Clause 5. They shall be marked with a type reference and identification of the VBPDST with which they are intended to be used.

The test pieces shall be opaque. Different test pieces can be required for different phases of the test procedures.

The characteristics of the test piece which shall be considered are:

- a) size;
- b) shape;
- c) colour;
- d) reflectivity;
 - at the wavelength of the illumination for VBPDEST with PAPT
 - at the wavelength of maximum sensitivity of the sensor for VBPDEST with AIT
- e) contrast with background;
- f) texture.

When defining the characteristics of the test piece, protection against camouflage with the background shall be taken into account.

4.2.13.2 Cylindrical test piece

The test piece shall be cylindrical if the VBPDEST is intended to be used for finger detection. The cylindrical test piece shall have a diameter of 14 mm and a length for ease of use.

4.2.13.3 Conical test pieces

The test piece shall be a truncated cone if the VBPDEST is intended to be used for hand detection. The test piece starts with a diameter of 20 mm increasing up to 40 mm over a length of 160 mm.

The test piece shall be a truncated cone in combination with a cylinder if the VBPDEST is intended to be used for arm detection. The test piece starts with a diameter of 40 mm increasing up to 55 mm as a cone over a length of 180 mm and continues as a cylinder with a diameter of 55 mm to an overall length of 440 mm.

The test piece shall be a truncated cone if the VBPDEST is intended to be used for leg detection. The test piece starts with a diameter of 50 mm increasing up to 117 mm over a length of 1 000 mm.

If the VBPDEST is intended to be used for detection of different parts of a body, the selection of the most appropriate test pieces shall be dependent on the analysis of the design and intended application. In some cases, all test pieces can be required.

4.2.13.4 Spherical test piece

If the VBPDEST is intended to be used for whole body detection, then the test piece shall be a sphere with a maximum diameter of 200 mm attached to a cylinder with a maximum diameter of 50 mm and a length selected for ease of use.

NOTE A spherical test piece with a diameter of 200 mm is intended to represent the thickness of a body.

4.2.13.5 Grey test piece (GTP)

The test piece shall have a diffuse reflectance value of 27 %-33 %.

4.2.13.6 Black test piece (BTP)

The test piece shall have a diffuse reflectance value of less than 5 %.

4.2.13.7 White test piece (WTP)

The test piece shall have a diffuse reflectance value of more than 70 %.

4.2.13.8 Retro-reflective test piece (RRTP)

The test piece shall have a retro-reflective surface that complies with the requirements for separate performance retro-reflective material of ISO 20471 or equivalent.

4.2.14 Wavelength

VBPDESTs shall operate at a wavelength within the range 400 nm to 1 500 nm.

4.2.15 Radiation intensity

If the emitter of the PAPT device uses LED technology, the radiation intensity generated and emitted by the VBPDEST shall meet the requirements of the exempt group in accordance with IEC 62471.

NOTE Exempt group is equal to risk group zero (IEC 62471).

If the emitter of the PAPT device uses laser technology, the radiation intensity generated and emitted by the VBPDEST shall at no time exceed the accessible emission limits (AEL) for a class 1M device in accordance with 4.3 of IEC 60825-1:2014.

4.2.16 Mechanical construction

When the detection capability can be decreased below the limit stated by the supplier as a result of a change of position of its components, the fixing of those components shall not rely solely on friction.

NOTE The use of oblong mounting holes without additional means could lead for example to a change of the position of the detection zone under mechanical interference such as bump.

4.3 Environmental requirements

4.3.1 Ambient air temperature range and humidity

Addition:

The ESPE shall not fail to danger when subjected to a rapid change of temperature and humidity leading to condensation on the optical window.

This requirement is verified by the condensing test of 5.4.2.

Additional environmental requirements:

4.3.5 Ambient light intensity

The VBPDEST shall continue in normal operation within a range of illumination on the background, limited from 100 lx to 1 500 lx at backgrounds defined in 5.1.2.4. If the supplier specifies limits outside this range, the tests shall be performed at the specified limits. Outside the limits, the VBPDEST shall not fail to danger.

4.3.6 Light interference

The VBPDEST shall continue in normal operation when subjected to the following:

- incandescent light;
- flashing beacons;

- fluorescent lights operated with high-frequency electronic and line power supply.

The VBPDST shall not fail to danger when subjected to:

- incandescent light;
- stroboscopic light;
- fluorescent lights operated with high-frequency electronic and line power supply;
- illumination fading to 0 lx;
- radiation from a VBPDST of identical design;
- collimated laser beam;
- direct sunlight.

These requirements shall be verified by the light interference test of 5.4.6.

The supplier shall inform the user of potential problems not covered by the requirements of this Technical specification.

Based on the technologies and algorithms used, and the analysis of 5.2.9, additional tests can be necessary.

4.3.7 Pollution interference

4.3.7.1 Effects on optical window

Pollution on the optical window shall not lead to a failure to danger.

Pollution resulting in a complete loss of the stated VBPDST detection capability shall cause the ESPE to go to a lock-out condition within the specified response time.

Pollution resulting in a deterioration of the stated VBPDST detection capability shall cause the ESPE to go to a lock-out condition within a time period of 5 s following the occurrence of the pollution interference.

4.3.7.2 Effects within the detection zone

Pollution within the detection zone or the zone with limited detection capability shall not lead to a failure to danger.

Pollution resulting in a complete loss of the stated VBPDST detection capability shall cause the ESPE to go to a lock-out condition within the specified response time.

Pollution resulting in a deterioration of the stated VBPDST detection capability shall cause the ESPE to go to a lock-out condition within a time period of 5 s following the occurrence of the pollution interference.

4.3.8 Manual interference

It shall not be possible to reduce the stated detection capability or modify the detection zone:

- by covering one or more of the optical windows of the housing of the VBPDST or other parts (if applicable);
- by placing objects within the zone with limited detection capability.

In such cases, the VBPDST shall respond by going to the OFF-state within a time period of 5 s and shall remain in the OFF-state until the manual interference is removed.

4.3.9 Optical occlusion (eclipsed by small object)

The VBPDST detection capability shall be maintained if moving or static objects or parts of a machine which are smaller than the detection capability and are present in the detection zone or in the zone with limited detection capability, which can block the view of the object to be detected. If the detection capability cannot be maintained, then the OSSDs shall go to the OFF-state and shall remain in the OFF-state until the optical occlusion is removed. This shall be verified by analysis and by a test according to 5.4.9.

NOTE Software filtering algorithms are sometimes provided to disregard small objects, for example, to increase reliability of operation.

4.3.10 Drift or ageing of components

Drift or ageing of components that would reduce the detection capability below the stated value shall not cause a failure to danger of the ESPE. The drift or ageing shall be detected within a time period of 5 s and lead to an OFF-state of the OSSDs.

If a reference object is used for monitoring ageing and drift of components, variations in the properties of the reference object (for example, reflectance) shall not cause a failure to danger of the ESPE. If a reference object is used to monitor ageing and drift of components, it shall be considered to be part of the VBPDST and shall be provided by the supplier of the VBPDST.

5 Testing

This clause of Part 1 is applicable except as follows:

5.1 General

5.1.2 Test conditions

5.1.2.1 Test environment

Replacement:

Except where otherwise specified in 5.4 of IEC 61496-1:2012, the tests shall be carried out with the ESPE operated under the following conditions:

- rated voltage (or a voltage within the rated voltage range);
- rated frequency (or a frequency within the rated frequency range);
- ambient temperature: 20 °C ± 5 °C;
- relative humidity: 25 % to 75 %;
- barometric pressure: 86 kPa to 106 kPa;
- homogeneous diffuse reflectivity of the background with no visible surface structure;
- where other non homogeneous surface characteristics are shown to be critical as result of the analysis of the design, a selection of critical backgrounds shall be applied.

NOTE Values stated in the marking and in the accompanying documents are considered as rated values.

Unless otherwise stated in this part of IEC 61496, the VBPDST shall be set up for the tests as follows:

- Typical contrast in accordance with 5.1.2.4;
- Ambient light intensity between 50 lx and 300 lx measured on the background.

The ambient light source should provide evenly distributed illumination as far as practical.

In the following tests, grey background (GB) is defined as a flat surface with 27 % to 33 % diffuse reflectivity measured at:

- the wavelength of the illumination for VBPDEST with PAPT;
- the wavelength of maximum sensitivity of the sensor for VBPDEST with AIT.

During the tests, the fixture holding the test piece should not be visible to the sensor (as much as practical).

5.1.2.2 Measurement accuracy

Addition to the first paragraph:

- for light intensity measurement: ± 10 % (of the value to be measured).

Addition:

5.1.2.4 Setup for typical test condition (TTC)

The test conditions shall be defined from the limits of detection capability specified in this part of IEC 61496 as a minimum or by the supplier, whatever is more stringent. Within these limits, the VBPDEST shall operate as specified by the supplier, outside these limits the VBPDEST shall not fail to danger.

The following setup shall be used:

- Grey background (GB);
- White test piece (WTP).

Ambient light intensity on the background within the range of 50 lx to 300 lx (TI).

5.1.2.5 Setup for low contrast (LC)

The following setup shall be used:

- Grey background (GB);
- Grey test piece (GTP).

Addition:

5.1.4 Test conditions and test plan

As a result of the analysis of the design and optical performance of the VBPDEST the test plan shall be established considering the test conditions and parameters outlined in this document.

The minimum test conditions shall be as specified in this technical specification or by the supplier, whichever is more stringent. Unless otherwise stated, the tests shall be done with the minimum detection zone positioned as specified in Table 1.

In the following tests, it shall be verified that when the OSSDs go to the OFF-state, they remain in the OFF-state while the test piece is present in the detection zone.

5.2 Functional tests

5.2.1 Sensing function

Addition:

5.2.1.1 General

The sensing function and the integrity of the detection capability shall be tested as specified, taking into account the following:

- The tests shall verify that the specified test pieces are detected when the test piece is either static or moving into or within the detection zone at any speed from 0 to 1,6 m/s . Where the supplier states that objects can be detected moving at higher speeds, the requirements shall be met at all speeds up to the stated maximum speeds.
- The tests shall verify that the specified test pieces are detected when the test piece is placed inside the stated detection zone(s) as far as the dimension of the stated detection capability.
- Tests shall be performed with the appropriate test piece inside the detection zone close to the background and close to the zone with limited detection capability and close to the tolerance zone(s). Tests at other locations can be required depending on analysis of the design and worst case considerations.
- For VBPDST with two imaging devices, the test piece shall be aligned parallel to the stereo base. This test is not applicable for the spherical test piece.
- The number, selections and conditions of the individual tests shall be such as to verify the requirements of 4.2.12.

If the detection capability depends on the direction of the movement, an analysis of the design and optical performance shall be done to identify the worst case conditions.

It shall be verified that the sensing device is continuously actuated and, where appropriate, that the OSSD(s) go to the OFF-state as described below, taking into account the operating principle of the VBPDST and, in particular, the techniques used to provide tolerance to environmental interference.

Table 1 shows an overview of the minimum tests required for the verification of detection capability requirements. Tests of Table 1 may be dismissed if the analysis of the VBPDST shows that they are covered by other tests that are more stringent.

Table 1 – Verification of detection capability requirements (see also 4.2.12)

Number	Test sequence	Sub clause	Test related to	Conditions	Distance between the sensing device and the test piece					
					Maximum operating distance		Middle operating distance		Minimum operating distance	
					Image centre ^a	Image corner ^a	Image centre ^a	Image corner ^a	Image centre ^a	Image corner ^a
1	B-Test	5.2.1.1	sensing function	Speed between 0 m/s and 1,6 m/s ^b	LC	LC	LC	LC	LC	LC
2	B-Test	5.2.1.1	sensing function	Speed between 0 m/s and 1,6 m/s ^b	GB, BTP	GB, BTP	GB, BTP	GB, BTP	GB, BTP	GB, BTP
3	B-Test	5.2.1.1	sensing function	Speed between 0 m/s and 1,6 m/s ^b	TTC	TTC	TTC	TTC	TTC	TTC
4	B-Test	5.2.1.1	sensing function	Speed between 0 m/s and 1,6 m/s ^b	GB, RRTP	GB, RRTP	GB, RRTP	GB, RRTP	GB, RRTP	GB, RRTP
5	Continuous B-Test	5.2.1.3	Endurance test				TTC			
6	C-Test	5.2.1.5	Periodic surface structures	Based on VBPDST specific analysis			TTC			

Number	Test sequence	Sub clause	Test related to	Conditions	Distance between the sensing device and the test piece					
					Maximum operating distance		Middle operating distance		Minimum operating distance	
					Image centre ^a	Image corner ^a	Image centre ^a	Image corner ^a	Image centre ^a	Image corner ^a
7	B-Test	5.4.2	Ambient temperature variation	50 °C or maximum ^c 5.4.2 of IEC 61496-1:2012 applies			TTC			
8	B-Test	5.4.2	Ambient temperature variation	0° C or minimum, non-condensing ^c 5.4.2 of IEC 61496-1:2012 applies			TTC			
9	C-Test	5.4.2	Humidity	5.4.2 of IEC 61496-1:2012 applies ^c			TTC			
10	B-Test	5.4.3 of IEC 61496-1:2012	Electrical disturbances	4.3.2, 5.2.3.1 and 5.4.3 of IEC 61496-1:2012 apply			TTC			
11	B-Test	5.4.4.1	Vibration	5.4.4.1 applies			TTC			
12	B-Test	5.4.4.2	Bump	5.4.4.2 applies			TTC			
13		5.4.6	Light interference	see Table 2						
14	B-Test	5.4.7	Pollution on the surface of the optical window (4.3.7.1)				TTC			
15	B-Test	5.4.7	Pollution in the detection zone (4.3.7.2)	Pollution caused by smoke and dust			TTC			
16	C-Test	5.4.8	Manual interference	Based on VBPDEST specific analysis			TTC			
17	B-Test or C-Test	5.4.9	Optical occlusion	Based on VBPDEST specific analysis			GB, BTP			
18		5.2.1.4	Tolerance zone	-		LC				

^a Location of image centre or image corner and the operating distance implies the location of the detection zone.

^b Or higher if specified by the manufacture

^c VBPDEST in test chamber and view through chamber window or detection zone for test in the chamber or open test chamber – start test within 1 min.

5.2.1.2 Integrity of the VBPDEST detection capability

It shall be verified that the VBPDEST detection capability is continuously maintained or the ESPE does not fail to danger by systematic analysis of the design of the VBPDEST, using testing where appropriate and/or required, taking into account 4.2.12.1.

5.2.1.3 Endurance test of the detection capability

It shall be verified that the detection capability is maintained by carrying out an endurance test as follows. The results of the analysis and testing according to 5.2.1.2 shall be used to determine the conditions and the appropriate test piece (see 4.2.13) to use for this test.

A limited functional test B (B test) in accordance with 5.2.3.3 of IEC 61496-1:2012 shall be carried out with the ESPE in continuous operation under the conditions determined for 96 h and with the test piece remaining in position inside the detection zone.

5.2.1.4 Position accuracy

The dimensions of the tolerance zone shall be verified in accordance with the requirements and results of 4.2.12.6 and 4.2.12.7. As a minimum, the test setup shall be in accordance with 5.1.2.5 (LC) at maximum operating distance with the test piece in the image corner.

Annex BB gives methods and information that can be used to verify tolerance zones.

5.2.1.5 Tests for the influence of periodic surface structures on the background

An analysis shall be carried out to determine whether periodic surface structures on the background (see Figure 4) or within the tolerance zone can influence the integrity of the detection capability.

The results of the analysis of the design regarding the influence of periodic surface structures on the detection capability shall be used to determine if the requirements of 4.2.12.9 can be met and to define details of the test setup.

The limitations of the system shall be identified taking into account the stated detection capability and worst case considerations regarding the characteristics of the pattern elements, including for example:

- size;
- shape;
- contrast.

The tests covering the influence of periodic surface structures on the detection capability shall be carried out using the following test setup procedure:

- a) Determination of the characteristics of the pattern elements depending on worst case considerations, especially taking into account the contrast detection and stereo algorithms.
- b) Test pieces: GTP, WTP, depending on worst case considerations;
- c) Test piece moving at any speed from 0 m/s to 1,6 m/s (or higher if specified by the supplier);
- d) The worst case distance between test piece and periodic surface structures on the background shall be determined by measurement.

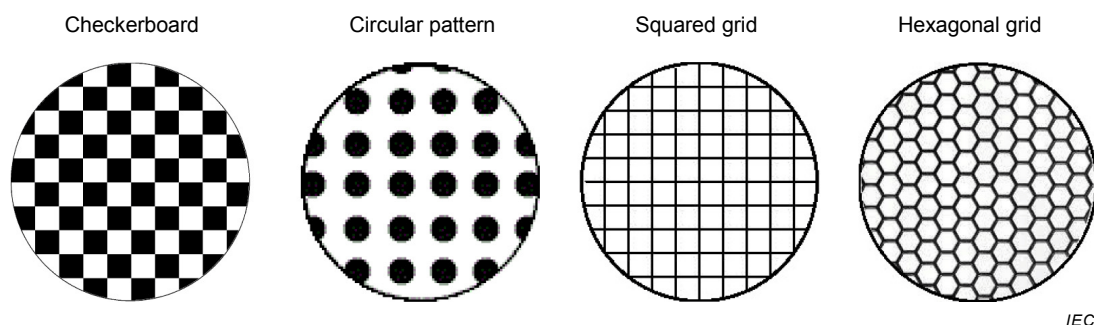


Figure 4 – Examples for periodic surface structures on the background

Additional subclauses:

5.2.9 Verification of optical performance

A systematic analysis of the electro-optical subsystem shall be carried out to:

- verify any filtering techniques (especially software filtering algorithms) employed, and their characteristics;
- determine the decision criteria used to determine whether or not the defined test piece(s) is detected as being inside the detection zone;
- determine the effect of undetected faults, in accordance with 4.2.2.4, on the electro-optical characteristics;
- determine worst case response time;
- determine the effect of environmental influence.

The results of this analysis shall be used to determine if the requirements of 4.1.2 can be met.

5.2.10 Wavelength

The wavelength used in the VBPDST shall be verified either by inspection of the device data sheets or by measurement.

5.2.11 Radiation intensity

If the emitter of the PAPT device uses LED technology, the radiation intensity shall be verified by measurement in accordance with IEC 62471 and inspection of the technical documentation provided by the supplier.

If the emitter of the PAPT device uses laser technology, the radiation intensity shall be verified by measurement in accordance with IEC 60825-1 and inspection of the technical documentation provided by the supplier. The marking as a class 1 or class 1M laser shall be verified for correctness.

5.3 Performance testing under fault conditions

5.3.2 Type 1 ESPE

This subclause of Part 1 is not applicable.

5.3.3 Type 2 ESPE

This subclause of Part 1 is not applicable.

5.3.4 Type 3 ESPE

Addition:

It shall be verified that the drift or ageing of components that influence the detection capability will be detected according to 4.3.10.

In practice it will be impossible to combine single faults with all operating conditions and/or influences listed in 4.2.12.1 by practical test. A combination of one or more of the following is sufficient to verify the requirement to combine single faults with operating conditions/influences as required by 4.2.2.4:

- analysis;
- simulation; and
- tests carried out in the presence of a single fault, where relevant.

5.3.5 Type 4 ESPE

This subclause of Part 1 is not applicable.

5.4 Environmental tests

5.4.2 Ambient temperature variation and humidity

Addition:

The VBPDST shall be subjected to the following condensation test:

- the VBPDST shall be supplied with its rated voltage and stored in a test chamber at an ambient temperature of 5 °C for 1 h;
- the ambient temperature and the humidity shall be changed within a time period of up to 2 min to a temperature of (25 ± 5) °C and a relative humidity of (70 ± 5) %;
- a C test shall be performed with the test piece remaining in the detection zone for a duration of 10 min using the TTC;
- if a restart interlock is available it shall not be operational during the C test.

As an alternative, a systematic analysis of the design of the electro-optical subsystem of the VBPDST may be carried out in order to justify the substitution of the condensing test by a test on the optical window (see 5.4.7).

5.4.4 Mechanical influences

5.4.4.1 Vibration

Addition:

If the sensing device of the VBPDST is not intended to be mounted on a machine (i.e. not intended to be subjected to high vibration), the levels of amplitude and frequency may be reduced for the A test depending on the intended application. In this case, a C test may be carried out instead of the B test.

At the end of the tests, the VBPDST shall be inspected for damage including displacement of optical components and mounting brackets. It shall be verified by test that the detection zone has not changed in orientation, size or position.

5.4.4.2 Bump

Addition:

If the sensing device of the VBPDST is not intended to be mounted on a machine (i.e. not intended to be subjected to bumps), the test conditions may be reduced for the A test depending on the intended application. In this case, a C test may be carried out instead of the B test.

At the end of the tests, the VBPDST shall be inspected for damage including displacement of optical components and mounting brackets. It shall be verified by test that the detection zone has not changed in orientation, size or position.

Additional environmental tests:

5.4.6 Light interference

5.4.6.1 General

Tests shall be performed with the test piece inside the detection zone close to the background and close to the zone with limited detection capability and close to the tolerance zone(s). Unless otherwise stated, the distance between the detection zone and the background shall be 2 times the dimension of the tolerance zone. Tests at locations other than image centre or image corner and with other test pieces may be required depending on analysis of the design and worst case considerations.

Each test shall be carried out as specified in Table 2 and under the stated conditions as a minimum requirement.

All the tests shall be performed with the minimum detection zone. The position of the detection zone shall be as specified in Table 2.

Additional tests shall be carried out under different combinations of operating distances and environmental conditions when

- the supplier states higher immunity levels, which shall be verified by testing at those levels with appropriate light sources, and/or
- an analysis shows such tests to be necessary.

Ambient light shall be delivered by using the incandescent light source or using natural illumination. Unless otherwise stated, the ambient light intensity during light interference tests shall be within a range of 50 lx to 300 lx.

In the following test procedures, unless otherwise stated, the light intensity limits include the combination of ambient light and light contributed by the indicated light source.

Table 2 gives an overview of the light interference tests.

Table 2 – Overview of light interference tests

Number	Sub-clause	Test related to	Light source position ^a Test sequence	Light intensity	Test piece position		Note
					Image centre ^b	Image corner ^b	
1	5.4.6.4	Normal operation Interference on background	Incandescent P1 1	1 500 lx	Max OD ^c TTC		Figure 5
2	5.4.6.4	Normal operation Interference on background	Flashing beacon P2 1		Max OD ^c TTC		Figure 5

Num- ber	Sub- clause	Test related to	Light source position ^a Test sequence	Light intensity	Test piece position		Note
					Image centre ^b	Image corner ^b	
3	5.4.6.4	Normal operation Interference on background	Incandescent with shadow P1 1	1 500 lx bright area ≤750 lx shadow area	Max OD ^c TTC	Max OD ^c TTC	Figure 5
4	5.4.6.6	Normal operation Interference on sensing device	Incandescent - 1	1 500 lx	Max OD ^c TTC	Max OD ^c TTC	Figure 7
5	5.4.6.6	Normal operation Interference on sensing device	Line frequency fluorescent - 1	750 lx	Max OD ^c TTC		Figure 7
6	5.4.6.6	Normal operation Interference on sensing device	High frequency fluorescent - 1	750 lx	Max OD ^c TTC		Figure 7
7	5.4.6.5	Failure to danger interference on background	Incandescent P1 2	3 000 lx	Max OD ^c LC		Figure 5
8	5.4.6.5	Failure to danger interference on background	Laser beam P1 2		Max OD ^c LC		Figure 5
9	5.4.6.5	Failure to danger interference on background	Direct sunlight on background - C test only	1x10 ⁵ lx	Analysis of the electro optical subsystem or test on component level		
10	5.4.6.5	Failure to danger Interference on background	Stroboscopic P2 2		Max OD ^c LC		Figure 5
11	5.4.6.7	Failure to danger Interference on sensing device	Laser beam - 2	Between 0,7 mW and 1 mW	Max OD ^c LC	Max OD ^c LC	Figure 7
12	5.4.6.7	Failure to danger Interference on sensing device	Incandescent - 2	3 000 lx	Max OD ^c LC	Max OD ^c LC	Figure 7
13	5.4.6.7	Failure to danger Interference on sensing device	Line frequency fluorescent - 2	3 000 lx	Max OD ^c LC	Max OD ^c LC	Figure 7
14	5.4.6.7	Failure to danger Interference on sensing device	High frequency fluorescent - 2	3 000 lx	Max OD ^c LC	Max OD ^c LC	Figure 7
15	5.4.6.7	Failure to danger Interference on sensing device	Stroboscopic 3 meter distance 3		Max OD ^c LC	Max OD ^c LC	Figure 7

Num- ber	Sub- clause	Test related to	Light source position ^a Test sequence	Light intensity	Test piece position		Note
					Image centre ^b	Image corner ^b	
16	5.4.6.7	Failure to danger Interference on sensing device	Flashing beacon 3 meter distance 3		Max OD ^c LC	Max OD ^c LC	Figure 7
17	5.4.6.8	Normal operation Low level light during B test	- - B test only	100 lx	Max OD ^c LC		
18	5.4.6.8	Failure to danger Fading light during C test	- - C test only	<=100 lx	Max OD ^c <=100 lx LC		
19	5.4.6.5	Failure to danger Interference on background	Emitting device of PAPT of identical design - 3	-	Max OD ^c LC		Figure 6
^a Position of the light sources (see Figure 5 or Figure 7) ^b Location of image centre or image corner and the operating distance implies the location of the detection zone ^c Max OD Maximum operating distance between the VBPDSST sensor and the test piece in the detection zone							

5.4.6.2 Light sources

The light sources shall be as follows:

- a) Incandescent light source: a linear tungsten halogen (quartz) lamp having characteristics within the following limits:
 - colour temperature: 3 000 K to 3 200 K;
 - input power: 500 W to 1 kW rated power;
 - rated voltage: any value within the range 100 V to 250 V;
 - supply voltage: rated voltage $\pm 5\%$, sinusoidal a.c. (50 Hz/60 Hz);
 - nominal length: 150 mm to 250 mm.
- b) Fluorescent light source: a linear fluorescent tube having characteristics within the following limits (operating without a reflector or diffuser):
 - size: T8 x 600 mm (26 mm nominal diameter);
 - rated power: 18 W to 20 W;
 - colour temperature: 3 000 K to 6 000 K;
 - operated at its rated supply voltage $\pm 5\%$, sinusoidal a.c. (50 Hz/60 Hz), or used in combination with an electronic ballast having an operating frequency within the range of 30 kHz to 40 kHz;
 - use of line or high frequency fluorescent light source depending on the analysis of the design and optical performance of the VBPDSST e.g.: frame rate, integration time, shutter.
- c) Flashing beacon light source: a light source employing a xenon flash tube (without enclosure, reflector or filter) having characteristics within the following limits:
 - flash duration: from 40 μ s to 1 200 μ s (measured to the half-intensity point);

- flash frequency: 0,5 Hz to 2 Hz;
 - input energy per flash: 3 joules to 5 joules;
- d) Stroboscopic light source: a stroboscope employing a xenon flash tube (without enclosure, reflector or filter) having characteristics within the following limits:
- flash duration: from 5 μ s to 30 μ s (measured to the half-intensity point);
 - flash frequency: 5 Hz to 200 Hz (adjustable range);
 - input energy per flash: 0,05 joule (at 200 Hz) to 0,5 joule (at 5 Hz).
- e) laser beam pointer: a collimated laser beam having characteristics within the following limits:
- flash duration: continuous wave mode;
 - wavelength: within 550 nm up to 670 nm;
 - beam shape: diameter below 5 mm;
 - light intensity: 0,7 mW up to 1 mW;
 - laser Class: 2.

5.4.6.3 Test sequences

NOTE The A, B, and C tests below are defined in IEC 61496-1:2012, 5.2.3

Test sequence 1:

- 1 – ESPE in normal operation
- 2 – Switch on interfering light
- 3 – B test
- 4 – Switch off ESPE for 5 s. Restore power. Reset start interlock, if fitted
- 5 – B test
- 6 – Switch off interfering light
- 7 – B test

Test sequence 2:

- 1 – ESPE in normal operation
- 2 – Switch on interfering light
- 3 – C tests repetitively for 1 min
- 4 – Switch off ESPE for 5 s. Restore power. Reset start interlock, if fitted
- 5 – C tests repetitively for 1 min
- 6 – Switch off interfering light
- 7 – C tests repetitively for 1 min

Test sequence 3:

- 1 – ESPE in normal operation
- 2 – Switch on the interfering light
- 3 – C tests repetitively for 3 min

If a system analysis proves that a switch-off cycle and a system reset lead to identical system behaviour, then the switch-off cycle may be replaced by a system reset.

5.4.6.4 Normal operation – Interference on background

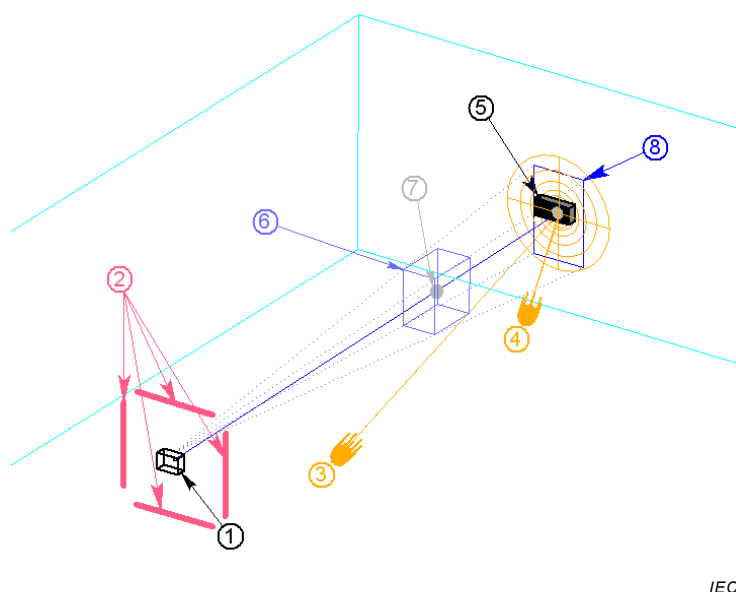
The ESPE shall continue in normal operation throughout test sequence 1 in 5.4.6.3 using light sources of 5.4.6.2, positioned outside the detection zone and illuminate the detection zone and the background.

Tests shall be carried out in accordance with Table 2.

Light intensity shall be measured according to Figure 5:

- the incandescent light source of 5.4.6.2 placed at P1 producing a light intensity of 1 500 lx. The test shall be performed with the incandescent light source beam moved slowly over the background through the optical axis and the corners of the detection zone;
- single incandescent light source of 5.4.6.2 placed at P2 with a spherical object held in front of the light source and outside the detection zone and the tolerance zone producing a shadow on the background surface. The size of the shadow shall be larger than the detection capability but less than 50 % of the projected area on the background of the minimum detection zone. On the background the light intensity shall be 1 500 lx in the bright area and ≤ 750 lx in the shadow;
- the flashing beacon light source of 5.4.6.2 shall be placed at P2 outside of the detection zone and the tolerance zone but at least at 2 m in height from the background;

If the supplier states higher immunity levels than 1 500 lx for this case, normal operation up to this level and no fail to danger above shall be verified by the test above.



Key

1 – VBPDSST	4 - Interfering light source P2	7 – Test piece
2 – Ambient light source	5 – Luxmeter	8 – Projected area
3 – Interfering light source P1	6 – Minimum detection zone	

Figure 5 – Test setup for indirect light interference on the background

5.4.6.5 Failure to danger – Interference on background

The ESPE shall not fail to danger throughout test sequences in accordance with Table 2 using light sources positioned outside the detection zone and the tolerance zone with the background illuminated.

Light intensity measurements shall be measured according to Figure 5.

- a) The incandescent light source of 5.4.6.2 placed at P1 producing a light intensity of 3 000 lx. The test shall be performed with the incandescent light source beam moved slowly over the background through the optical axis and the corners of the detection zone.
- b) The stroboscopic light source of 5.4.6.2 shall be placed at P2 outside of the detection zone and the tolerance zone but at least at 2 m in height from the background.
- c) The laser beam source of 5.4.6.2 placed at P1 producing a laser beam with a maximum diameter of 5 mm on the background. The test shall be performed with the laser beam moved slowly over the background through the optical axis and the corners of the detection zone;
- d) The VBPDEST with PAPT shall not fail to danger in the presence of a PAPT emitting device of identical design (see Figure 6);
- e) The VBPDEST shall not fail to danger by direct sunlight on the background (10^5 lx). This requirement shall be verified by:
 - the analysis of the electro optical subsystem considering 4.2.12.1;
 - or by testing at those levels with appropriate light sources.

NOTE Light intensity values are based on values given by EN 12464-1. The position and direction of the luxmeter is limited to achieve a reproducible light intensity value.

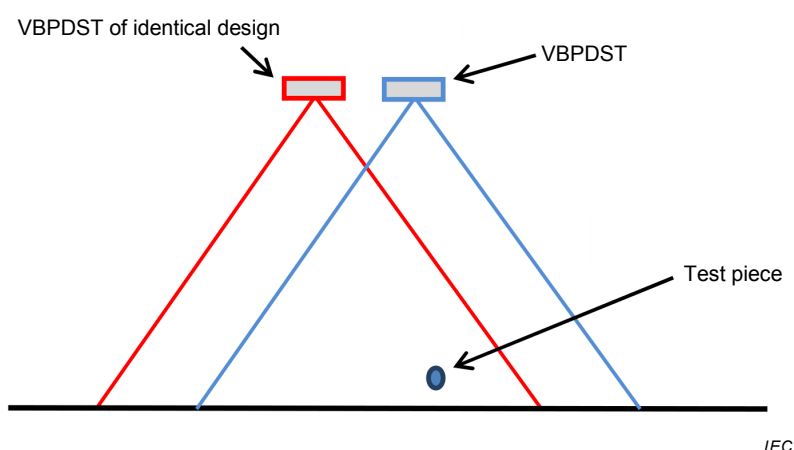


Figure 6 – Test setup for VBPDEST of identical design with PAPT

5.4.6.6 Normal operation – Interference on sensing device

The ESPE shall continue in normal operation throughout test sequence 1 in 5.4.6.3 using the light sources of 5.4.6.2.

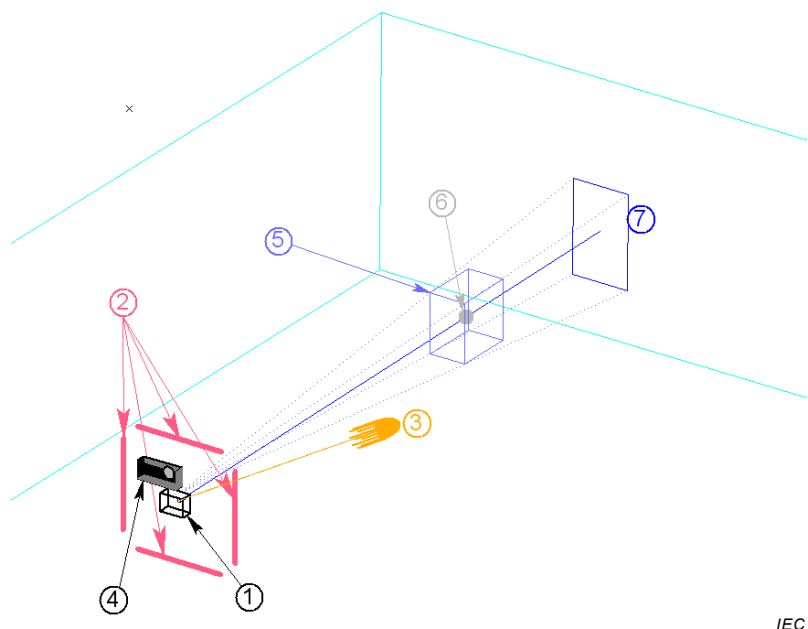
Tests shall be carried out in accordance with Table 2. The system shall be configured with the minimum detection zone located in the corner of the detection zone.

Light intensity measurements shall be made on the sensing device and perpendicular to the optical axis.

- a) The incandescent light source of 5.4.6.2 producing a light intensity of 1 500 lx and illuminating the sensing device adjacent to the field of view of all imaging sensors.
- b) The line frequency fluorescent light source of 5.4.6.2 producing a light intensity of 750 lx and illuminating the sensing device adjacent to the tolerance zone but outside the detection zone.
- c) The high frequency fluorescent light source of 5.4.6.2 producing a light intensity of 750 lx and illuminating the sensing device adjacent to the tolerance zone but outside the detection zone.

The ESPE shall not go to the ON-state when the test sequence requires it to be in the OFF-state.

The luxmeter shall be positioned on the sensing device and perpendicular to the optical axis.



Key

1 – VBPDS	4 – Luxmeter	7 – Projected area
2 – Ambient light source	5 – Minimum detection zone	
3 – Interfering light source	6 – Test piece	

Figure 7 – Test setup for direct light interference on the sensing device

5.4.6.7 Failure to danger – Interference on sensing device

The ESPE shall not fail to danger throughout test sequences in accordance with Table 2 using each of the light sources of 5.4.6.2 positioned outside the detection zone and adjacent to the tolerance zone, illuminating the sensing device.

Light intensity measurements shall be made on the sensing device and perpendicular to the optical axis.

5.4.6.8 Low lighting level test

With the VBPDS in normal operation, a B test shall be carried out with an ambient light intensity of 100 lx.

With the VBPDS in normal operation, a C test shall be carried out while the intensity of ambient light is decreased until the OSSDs go to the OFF-state.

5.4.7 Pollution interference

A systematic analysis of the design of the VBPDS shall be carried out to decide which tests (if any), test methods and test conditions are appropriate to satisfy the requirements of 4.3.7. These tests shall verify no failure to danger.

Examples of possible pollution are:

- dust, smoke, steam, spray, etc. in front of the sensing device;
- polluted window by particle, water-drops or dust.

5.4.8 Manual interference

Immunity against coverage shall be tested as follows:

- The objects used for simulating coverage shall be a circular opaque test spot with the diameter of 10 mm and a second test spot with the size and shape to cover the complete optical window of imaging device and emitter of the PAPT device if applicable.
- During the tests the spot shall be placed consecutively on each optical window.
- For systems where the optical windows (or other relevant parts) have a diameter larger than 10 mm, the test spot shall be placed at a position most relevant to the detection capability of the VBPDEST.
- Test for each covered part of the VBPDEST whether the simulated manual interference will lead to an OFF-state of the OSSDs within a time period of 5 s or does not reduce the stated detection capability.
- Tests shall be carried out to verify that when simulated manual interference leads to an OFF-state of the OSSDs, actuation of the restart interlock (if applicable) or a new power-up does not lead to an ON-state of the OSSDs. If a restart interlock is fitted, the OSSDs shall stay in the OFF-state when the covering test spot is removed.

5.4.9 Optical occlusion

Optical occlusion on the window, in the zone with limited detection capability or in the detection zone shall not reduce the stated detection capability. These tests shall be carried out to test for failure to danger.

Immunity against the effect of optical occlusion within the detection zone shall be tested as follows:

- a) The object used for simulating optical occlusion shall be a cylinder with a minimum effective length of 0,3 m. The diameter of the occluding object shall be 5 mm unless determined otherwise by the analysis of 4.3.9. The surface of the occluding object shall have a diffuse reflectance value of less than 20 % at:
 - the wavelength of the illumination for VBPDEST with PAPT;
 - the wavelength of maximum sensitivity of the sensor for VBPDEST with AIT.
- b) During the test, the occluding object shall be used normal to the optical axis of one of the imaging devices.
- c) The detection zone shall range from the minimum to the maximum operating distance, when applicable.
- d) The test shall be carried out by placing the occluding object in the detection zone as near as possible to the VBPDEST and in the zone with limited detection capability unless determined otherwise by the analysis of 4.3.9. A black test piece (see 4.2.13.6) and grey background shall be used. If the occluding object does not lead to the OFF-state of the OSSDs, a B test shall be performed otherwise a C test shall be performed.
- e) C-tests shall be performed to verify that the stated detection capability is maintained in the presence of optical occlusion. The black test piece shall be moved through the optical occlusion as close as possible to the occluding object and at the stated maximum detection distance.
- f) Additional tests shall be carried out when the analysis of 4.3.9 shows that the following can affect the immunity to optical occlusion:
 - 1) distances between the VBPDEST and the occluding object other than those stated above;
 - 2) dimensions of the detection zone other than the maximum;

- 3) other distances between the occluding object and the test piece;
- 4) different diameters of the occluding object at different distances from the VBPDST;
- 5) different positions of the occluding object in front of the VBPDST;
- 6) and/or more than one occluding object.

6 Marking for identification and for safe use

This clause of Part 1 is applicable except as follows:

6.1 General

Addition:

The markings required by 6.1 b), c) and d) of IEC 61496-1:2012 may alternatively be given in the accompanying documents.

7 Accompanying documents

This clause of Part 1 is applicable except as follows.

Addition:

The accompanying documents shall contain the following information where applicable:

- aaa) application examples showing the tolerance zone(s);
- bbb) dimensions of maximum and minimum detection zone(s) and tolerance zone(s) together with information about a reference point relative to the sensing device for the determination of the detection zone;
- ccc) information about the minimum required distance between the border of a detection zone and the surrounding environment without detecting, for example, walls or parts of machines in order to ensure reliability in operation;
- ddd) instructions for setting the detection zone(s) including consideration of the tolerance zone(s) and details on other optional functions of the VBPDST, described in Annex A of this part if these options are available. A clear statement shall be given when one or more zone(s) is(are) described, whether its(their) description is related to the detection zone(s) as defined in 3.4 or the combination of the detection zone(s) and the tolerance zone(s);
- eee) information about the behaviour of the VBPDST in the presence of smoke and dust;
- fff) information on how the detection capability can be affected if the VBPDST is used within an additional housing. For example, additional housings may have an influence on the detection capability and the detection zone;
- ggg) if appropriate for the application(s), an indication on the floor of the detection zone should be recommended;
- hhh) instructions on how to document on paper the setting of the detection zone(s) together with date, serial number of the VBPDST and identification of the person responsible;
- iii) instructions to follow when the VBPDST can be influenced during normal operation by a VBPDST of identical design;
- jjj) potential problems not covered by the requirements of this Technical Specification; especially information concerning external influences which are not covered by this Technical Specification and which can decrease the stated detection capability. Examples can include weld splatter, infra-red remote control devices, different fluorescent and stroboscopic light sources, snow, rain, pollution and thermal convection;

- kkk) relevant information concerning the need to check periodically the optical window(s) for damage (depending on the application);
- III) relevant information concerning the need to check periodically the mounting of the VBPDEST for correctness and to check for possible misalignment of the detection zone(s) (depending on the device and the application);
- mmm) information as required by 4.1.6 if the VBPDEST possesses one or more zone(s) with limited detection capability;
- nnn) information regarding the maximum speed in the worst case direction within the detection zone of the VBPDEST of an object having the minimum detectable size (see 4.2.12.5);
- ooo) instruction that the detection capability dimension shall be added to the safe distance calculations of ISO 13855. This is because response time specifications assume that the object is entirely within the detection zone before it is detected.

Annex A (normative)

Optional functions of the ESPE

Annex A of Part 1 applies except as follows.

Clause A.8 does not apply.

Additional optional functions:

A.9 Setting the detection zone and/or other safety-related parameters

A.9.1 Functional requirements

The setting of the detection zone and/or other safety-related parameters shall not be possible without using a key, keyword or tool.

NOTE For example, the tool can be a password protected software configuration program that is part of the VBPDST.

If the setting is carried out using a personal computer or equivalent fitted with untested dedicated hardware and/or software, a special procedure shall be used for setting the detection zone. This procedure shall be in accordance with appropriate standards (see also 4.2.11 of IEC 61496-1:2012). If the tool is software, only software authorized by the supplier shall be used.

The procedure shall include measures to ensure that the input parameters are transmitted correctly and without corruption to the VBPDST. This shall be applied for all safety related settings, for example, the setting of the response time. The parameterization procedures shall conform to an appropriate standard (e.g. IEC 62061:2005, 6.11.2 or ISO 13849-1:2006, 4.6.4).

The setting of safety-related parameters should only be performed by qualified persons.

A.9.2 Verification

Verify by inspection or test that:

- a) the setting function(s) for each configuration parameter (minimum, maximum and representative values) work(s) correctly;

NOTE It is possible that the detection zone displayed on the screen of a configuration tool (for example, a personal computer) can be different from the actual detection zone of the ESPE.

- b) the configuration parameters are checked for plausibility, for example by use of invalid values, etc.;
- c) the access to, and methods of, configuration by the user are in accordance with the requirements of corresponding standards (see, for example, 4.2.11 of IEC 61496-1:2012, or other relevant standards);
- d) in the case of detection zones that can be varied in size during operation, that the data/signals for determining the size of a detection zone are generated and processed in such a way that a single fault shall not lead to a loss of the safety function. And that such a single fault is detected and causes the OSSDs to remain in the OFF-state or to go to the OFF-state within the response time of the ESPE.

A.10 Selection of multiple detection zones

A.10.1 Functional requirements

If a VBPDST has more than one safety-related detection zone, a single fault shall not lead to an unintended change from one selected zone to another zone. In cases where a single fault which does not cause a failure to danger of the VBPDST is not detected, the occurrence of further faults internal to the VBPDST shall not cause a failure to danger.

Where the input signals are derived from device(s) external to the VBPDST, the latter should meet the relevant requirements of other appropriate standards (for example ISO 13849-1, IEC 61508, IEC 62061).

Single faults that prevent an intended change from one selected zone to another or prevent the activation of an additional safety-related detection zone shall cause the VBPDST to go to a lock-out condition when a demand requires an activation of another zone or an activation of an additional zone. The specified response time(s) shall be maintained in this case.

NOTE 1 It is possible that each zone has a different response time as specified by the manufacturer.

If a detection zone is changed in size on-line for example by external inputs, the same requirement applies.

The activation of the detection zones shall be monitored by the VBPDST. The user shall have the possibility to configure the sequence of activation of the detection zones which is monitored by the VBPDST. If an incorrect sequence of activation of the detection zones is detected, the VBPDST shall respond by going to a lock-out condition.

The possibility that persons may already be within a detection zone at the moment of switching between different detection zones should be considered.

NOTE 2 The automatic selection of safety-related detection zones is not a muting function (as described in Clause A.7 of IEC 61496-1:2012).

A.10.2 Verification

Verify by inspection or test that:

- a) a single fault does not lead to an unintended change from one selected zone to another zone; a single fault does not prevent an intended change from one selected zone to another or prevent the activation of an additional safety-related detection zone; a further fault will not lead to a failure to danger (according to 5.3.4).
- b) common-mode failures cannot lead to a deactivation or variation of the detection zones.
- c) the specified response time of the VBPDST is maintained in the case of switching between different detection zones.
- d) the user has the possibility to configure the sequence of activation of the detection zones which is monitored by the VBPDST.
- e) the VBPDST goes to the lock-out condition when the sequence of activation differs from that configured by the user.

Annex B (normative)

Catalogue of single faults affecting the electrical equipment of the ESPE, to be applied as specified in 5.3

Annex B of Part 1 is applicable except as follows.

Addition:

B.7 Imaging sensor

Faults considered	Exclusions
Wrong line addressing	None
Wrong column addressing	None
Crosstalk between lines, columns and pixels	None
Static image (no new image)	None
Stuck at high pixel	None
Stuck at low pixel	None
Change in register settings, if applicable	None
Failure in the analog to digital converter, if applicable	None
Failure in data pre-processing, if applicable (see Figure 1)	None

Annex AA (informative)

The positioning of VBPDST employing a volume as a detection zone in respect of parts of the human body

AA.1 Calculation of distances for electro-sensitive protective equipment employing vision based protective devices (VBPDST)

AA.1.1 General

NOTE 1 ISO 13855 provides a methodology to determine the minimum distance S from specific sensing or actuating devices of protective equipment to a danger zone. Clause 6 of ISO 13855:2010 details the calculation of minimum distances for electro-sensitive protective equipment employing active opto-electronic protective systems. This Annex AA adopts the given approach and extends it where necessary. It is foreseen that after some experience, the methodology will be presented to the committees preparing ISO 13855 and IEC 62046 for adoption and integration in their standards.

When calculating minimum distances, the requirements and formulae given by ISO 13855:2010, Clause 6 should be taken into account including additions given by AA.1.2 to AA.1.4.

ISO 13855 distinguishes in the calculation of the minimum distance between:

- detection zone orthogonal to the direction of approach; and
- detection zone parallel to the direction of approach.

Both cases can be applied for a three-dimensional volume; it is allowed to choose the resulting lower minimum distance S . Analysis has shown that the formulae for detection zones orthogonal to the direction of approach lead to a lower or equal minimum distance S in the cases described below. For the Formulae (AA.5) to (AA.9) it is considered that the outer shell of the three-dimensional detection zone is normal to the reference plane, e.g. floor. Other shapes such as ball-shaped or trapezoidal need further consideration. In addition, possible circumventing of a VBPDST by reaching over the detection zone has to be addressed according to ISO 13855:2010, Table 1.

To ensure that the value C_{RO} according to ISO 13855:2010, Table 1 is smaller than the value of $(C + d)$ calculated according to the formulae below independent of the height a of the hazard zone the height b of the upper edge of the detection zone of the VBPDST should be 1 400 mm as a minimum for a detection capability ≥ 70 mm and 2 400 mm as a minimum in all other cases.

NOTE 2 A height b equal or greater than 1 400 mm respectively 2 400 mm means that there is no possible circumventing of an ESPE according to ISO 13855 by reaching over the detection zone.

AA.1.2 Calculation of the overall minimum distance S_o

When calculating the size or volume of a zone that is used to prevent a person reaching the hazard zone before the termination of the hazardous machine function, an overall minimum distance S_o should be calculated by Formulae (AA.1) or (AA.4) as is appropriate. Formula (AA.2) is a general formula given by ISO 13855.

$$S_o = S + S_a \quad (\text{AA.1})$$

$$S = (K \times T) + C \quad (\text{AA.2})$$

$$S_a = C_{tz} + d \quad (\text{AA.3})$$

$$S_o = (K \times T) + C + C_{tz} + d \quad (\text{AA.4})$$

where:

- S_o is the overall minimum distance, in millimetres, combining the minimum distance S and an additional distance S_a ;
- S is the minimum distance, in millimetres, from the hazard zone to the detection point, line, plane or zone;
- S_a is an additional distance, in millimetres, combining the effects of systematic and random influences;
- K is a parameter in millimetres per second, derived from data on approach speeds of the body or parts of the body (see ISO 13855 for details);
- T is the overall system stopping performance in seconds;
- C is an additional distance in millimetres, based on the distance, which a part of the body may be moving towards the hazard zone prior to the actuation of the protective device;
- C_{tz} is an additional distance in millimetres, based on the tolerance zone of the protective device to satisfy systematic and random influences;
- d is the sensor detection capability of the device, in millimetres (mm), i.e. the dimension of the test piece.

NOTE Protective devices employing a volume as a detection zone will normally require a test piece to be inside the detection zone with a dimension of at least its stated detection capability. This is taken into account by the corresponding test procedures (see for example Clause 5). Therefore the dimension of the test piece (d) is part of the additional distance S_a in the Formulae (AA.3) and (AA.4). If partial intrusion (see AA.1.5, Example 2) satisfies the requirements of this part of IEC 61496, then only the relevant portion of the dimension d is used in those formulae (i.e. d_1 in Figure AA.4 to Figure AA.6).

AA.1.3 Vision based protective devices with a detection capability > 40 mm and ≤ 55 mm

The minimum distance S in millimetres should be calculated by Formula (AA.5) for VBPDST having a detection capability in the range > 40 mm and ≤ 55 mm

$$S = (K \times T) + C_{40} + C_{55} \quad (\text{AA.5})$$

where:

- S is the minimum distance, in millimetres, from the hazard zone to the detection point, line, plane or zone;
- K is a parameter in millimetres per second, derived from data on approach speeds of the body or parts of the body (see ISO 13855 for details);
- T is the overall system stopping performance in seconds;
- C_{40} is an additional distance in millimetres, based on formulae given by ISO 13855 with $C_{40} = 8 (d - 14 \text{ mm}) = 8 (40 \text{ mm} - 14 \text{ mm}) = 208 \text{ mm}$;
- C_{55} is an additional distance in millimetres, based on the formula $C_{55} = 12 (d - 40 \text{ mm})$;
- d is the sensor detection capability of the device, in millimetres (mm), i.e. the dimension of the test piece.

Then

$$S = (K \times T) + 208 \text{ mm} + 12 (d - 40 \text{ mm}) \quad (\text{AA.6})$$

$$S = (K \times T) + 12 d - 272 \text{ mm} \quad (\text{AA.7})$$

NOTE 1 The formula for C_{55} is derived from an estimation based on data given by B. Flügel, H. Greil, K. Sommer, Anthropologischer Atlas, Verlag Tribüne Berlin 1986, ISBN 3-7303-0042-3.

NOTE 2 For the calculation of C_{40} the value of d is 40 mm irrespective of the detection capability stated by the manufacturer. For the calculation of C_{55} the value of d is the detection capability stated by the manufacturer.

AA.1.4 Vision based protective devices with a detection capability > 55 mm and ≤ 200 mm

The minimum distance S in millimetres should be calculated by Formula (AA.8) for VBPDSST having a detection capability in the range > 55 mm and ≤ 200 mm.

$$S = (K \times T) + C \quad (\text{AA.8})$$

where:

S is the minimum distance, in millimetres, from the hazard zone to the detection point, line, plane or zone;

K is a parameter in millimetres per second, derived from data on approach speeds of the body or parts of the body with $K = 1\,600$ mm/s (see ISO 13855 for details);

T is the overall system stopping performance in seconds;

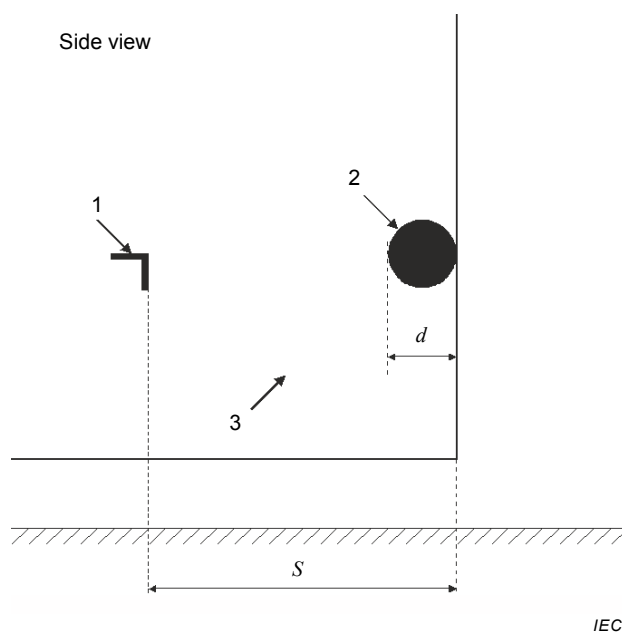
C is an additional distance of 850 mm; according to ISO 13855 this value is considered to be the standard arm reach.

Then see (AA.9):

$$S = (1\,600 \text{ mm/s} \times T) + 850 \text{ mm} \quad (\text{AA.9})$$

AA.1.5 Examples of detection zone and tolerance zone

For description of abbreviations used in Figure AA.1 to Figure AA.6, see AA.1.2.



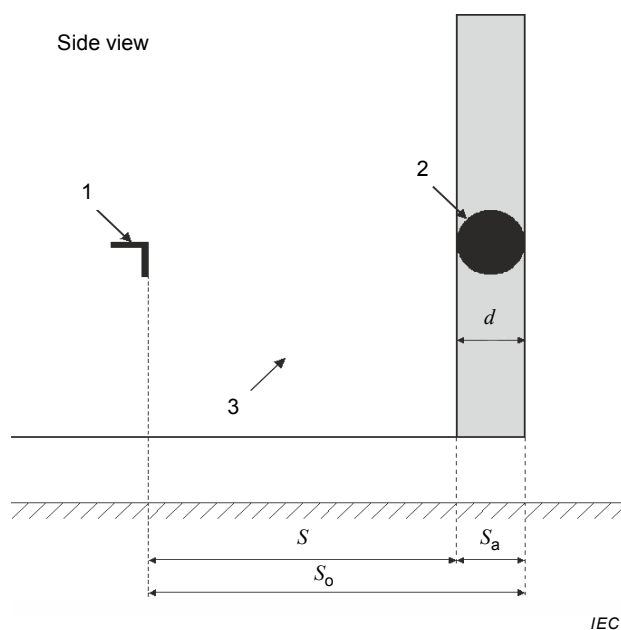
IEC

Key

- 1 – Hazard zone
- 2 – Test piece
- 3 – Detection zone

Figure AA.1 – Minimum distance S – Example 1

According to the general description of the test procedure in 5.2.1.1, the test piece shall be detected when placed inside the detection zone as far as the stated detection capability d .

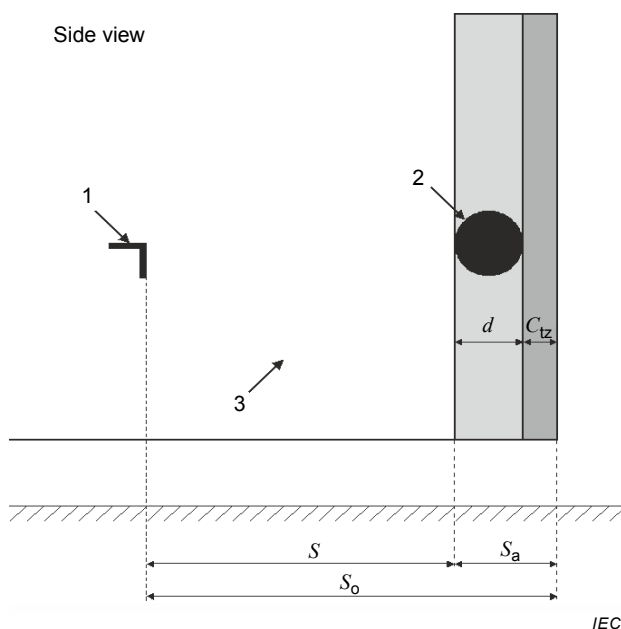


Key

- 1 – Hazard zone
- 2 – Test piece
- 3 – Detection zone

Figure AA.2 – Overall minimum distance S_o without tolerance zone – Example 1

The dimension of the detection capability d shall be added to the minimum distance S to ensure the correct distance between the hazard zone and an object.

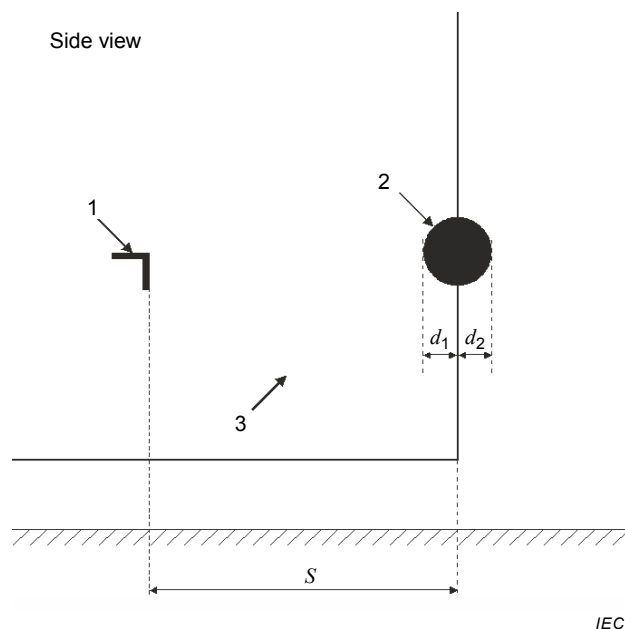


Key

- 1 – Hazard zone
- 2 – Test piece
- 3 – Detection zone

Figure AA.3 – Overall minimum distance S_o including tolerance zone – Example 1

To achieve the required minimum probability of detection, the tolerance zone has to be considered in addition to the detection zone. The tolerance zone depends on position accuracy composed of systematic (STZ) and random influences (PTZ). Even if a measured distance value of a test piece falls into the tolerance zone, this test piece will be determined as detected and the OSSDs will go to the OFF-state or remain in the OFF-state.

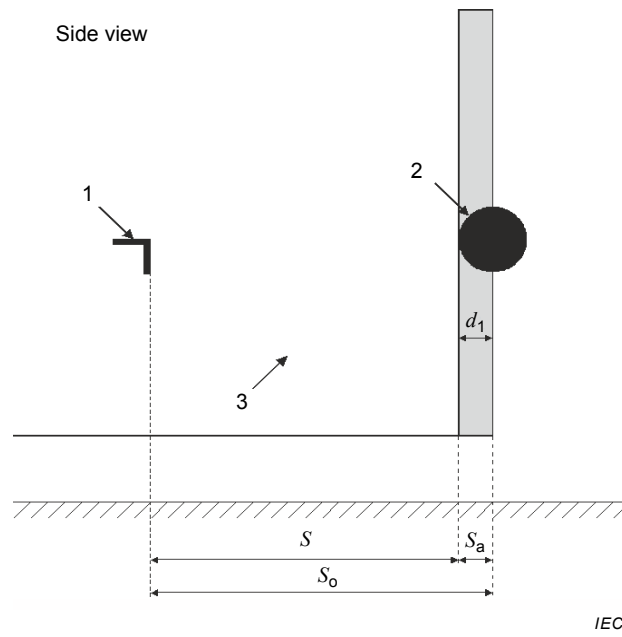


Key

- 1 – Hazard zone
 - 2 – Test piece
 - 3 – Detection zone
- $d = d_1 + d_2$

Figure AA.4 – Minimum distance S – Example 2

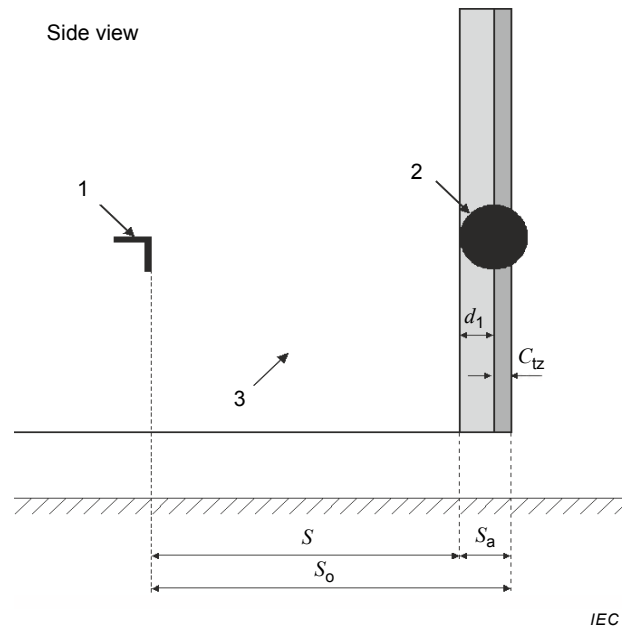
According to the general description of the test procedure in 5.2.1.1 the test piece shall be detected when placed inside the detection zone as far as the stated detection capability d . If partial intrusion of an object into the detection zone as shown by the dimension d_1 in Figure AA.4 leads to detection, then the dimension d_1 shall be added to the minimum distance S to ensure the correct distance between the hazard zone and an object, see Figure AA.5.



Key

- 1 – Hazard zone
- 2 – Test piece
- 3 – Detection zone
- d_1 – see Figure AA.4

Figure AA.5 – Overall minimum distance S_0 without tolerance zone – Example 2



Key

- 1 – Hazard zone
- 2 – Test piece
- 3 – Detection zone
- d_1 – see Figure AA.4

Figure AA.6 – Overall minimum distance S_0 including tolerance zone – Example 2

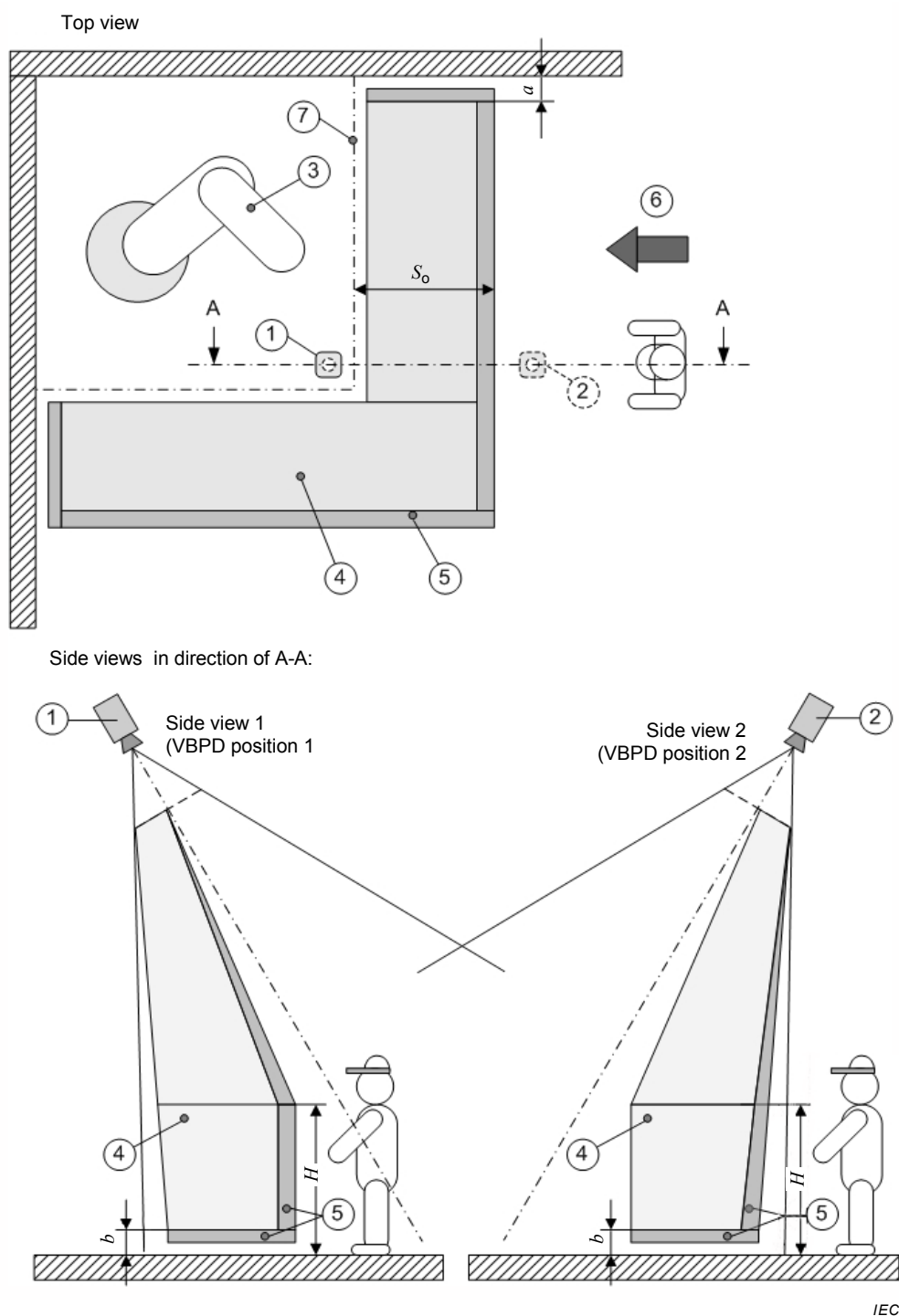
To achieve the required minimum probability of detection, the tolerance zone has to be considered in addition to the detection zone. The tolerance zone depends on position accuracy composed of systematic (STZ) and random influences (PTZ). Even if a position of a test piece falls into the tolerance zone, this test piece will be determined as detected and the OSSDs will go to the OFF-state or remain in the OFF-state.

AA.2 Application examples for body detection of a VBPDST employing a volume as a detection zone

The application example (see Figure AA.7) shows a robot working station. The entry of persons is prevented on two sides by walls or safety fences (top view). The entry from the other two sides is monitored by a VBPDST. If a person enters the detection zone, the robot stops its movement.

The safety distance S_0 should be calculated in accordance with the examples given in AA.1.5 and the VBPDST accompanying documents. Furthermore, the distance to the walls (a), the distance to the floor (b) and the height of the detection zone (H) should be calculated in accordance with the VBPDST accompanying documents and the risk assessment.

The VBPDST can be mounted at different positions. Side view 1 and side view 2 show two possible solutions and the shape of the resulting detection zone (4) and the adjacent tolerance zone (5).



Key

1 – VBPDS position 1

2 – VBPDS position 2

3 – robot

4 – detection zone

5 – tolerance zone S_a

6 – direction of approach

7 – hazard zone

S_o – Safety distance according AA.1.2

H – height of detection zone over the floor level

a – distance to walls

b – distance to the floor

NOTE The dimensions of a and b are according to the risk assessment.

Figure AA.7 – Application example for body detection of a VBPDS employing a volume as a detection zone

Annex BB (informative)

Relationship between position accuracy and tolerance zones for VBPDST

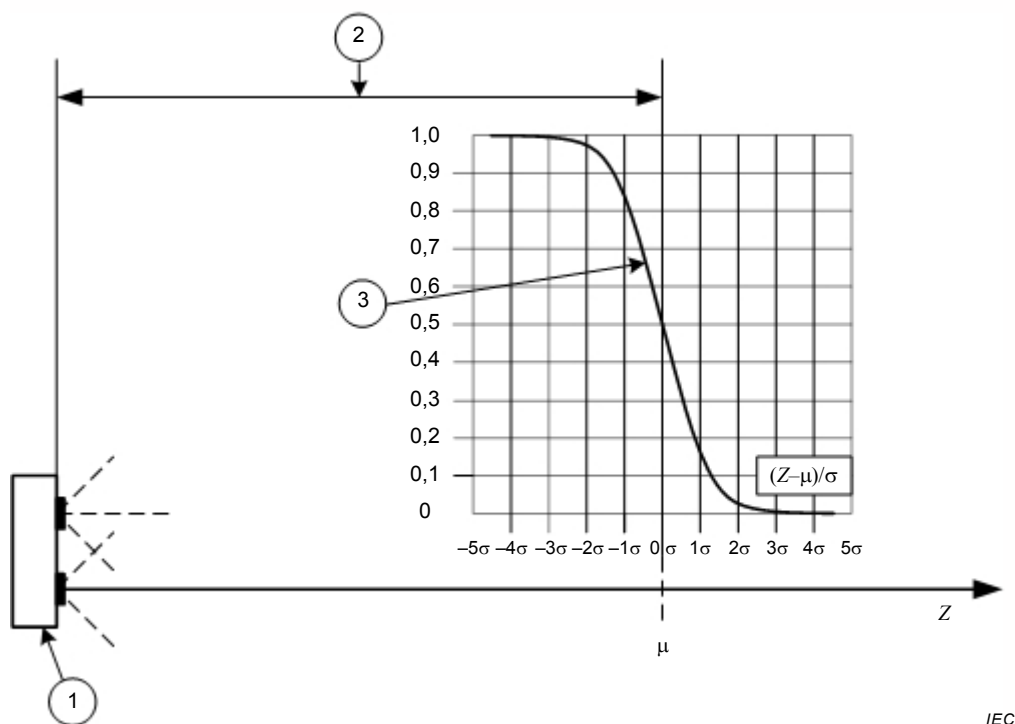
BB.1 Probability of detection

Probability of detection as used in this Technical Specification is determined by the accuracy of measurement and is not related to the probability of faults. The probability with distribution function F_D that a test piece placed at the border of the detection zone is measured as being inside the detection zone can be calculated by using the normal distribution function F per Formula (BB.1).

$$F_D(\mu) = 1 - F(\mu) = 1 - \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{1}{2}\left(\frac{z-\mu}{\sigma}\right)^2} dz = 0,5 \quad (\text{B.1})$$

This calculation is based on the assumption that the measurement values follow a normal (Gaussian) distribution.

One method to measure the probability of detection is to use the detection information of the VBPDST (ON or OFF state of the OSSDs) for systems not providing object distance information. Figure BB.1 shows the relationship between test piece position and the probability of detection in one direction. Other directions may have to be considered accordingly.



Key

- 1 – Sensing device
- 2 – Detection zone
- 3 – Probability of detection

Figure BB.1 – Relationship between test piece position and the probability of detection

According to the relationship shown in Figure BB.1, a test piece will be measured as inside the detection zone with a probability of 0,5 at the position $z = \mu$. Without any addition to the detection zone, the probability of detection would be unacceptably low. It is a requirement of this Technical Specification that the supplier states an addition which is called the tolerance zone. Several different influences contribute to the tolerance zone as defined in this Technical Specification.

BB.2 Tolerance zone related to probability

One part of the tolerance zone S_a is related to probability (PTZ). The probability that a test piece is measured as being inside the detection zone can be calculated according to Formula (BB.2).

$$F_D(\mu - 5\sigma) = 1 - \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu-5\sigma} e^{-\frac{1}{2}\left(\frac{z-\mu}{\sigma}\right)^2} dz = 1 - 2,9 \times 10^{-7} \quad (\text{BB.2})$$

This calculation is based on the assumption that the measurement values follow a normal (Gaussian) distribution. When configuring the detection zone, a tolerance zone related to the deviation of 5σ should be added to the calculated safety distance (detection zone). Then the probability that it will be measured as inside the detection zone is $1 - 2,9 \times 10^{-7}$.

BB.3 Determination of tolerance zone for systems not providing object distance information

A practical way to measure σ is to measure the probability of detection at different positions z_i near to the outer border of a fix detection zone under worst case conditions and to recalculate σ by differentiation of the discrete probability of detection values $F(z_i)$ (as shown in Figure BB.2).

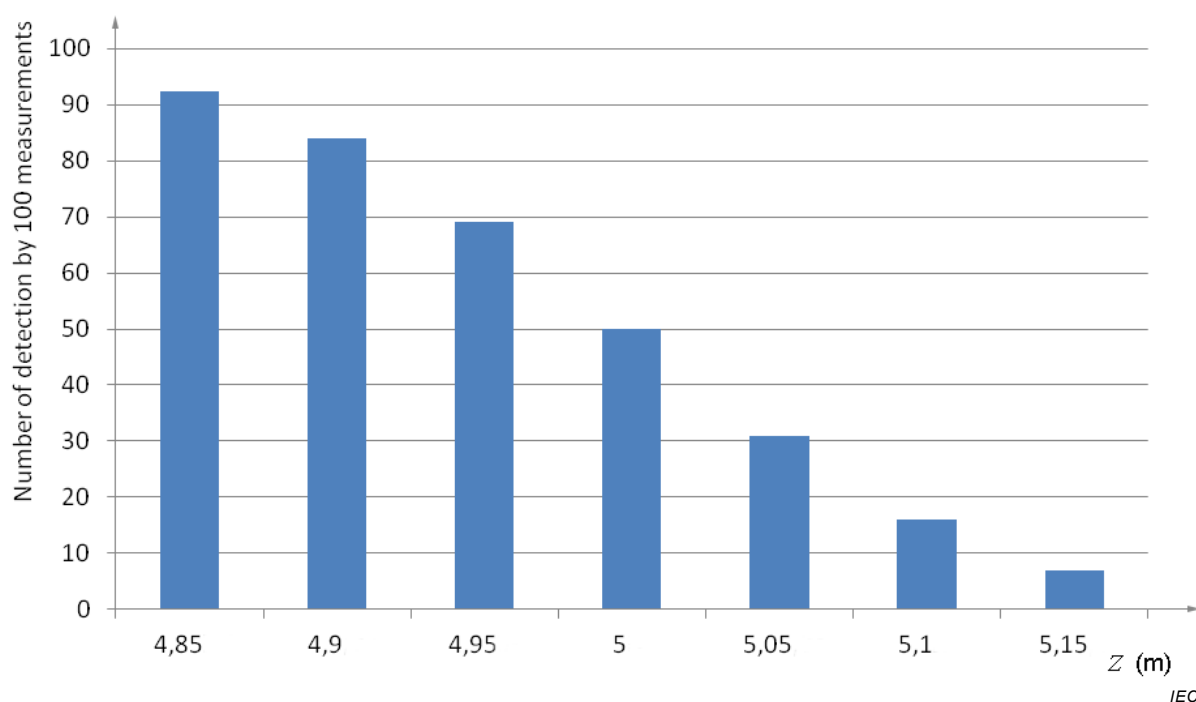


Figure BB.2 – Example for measurement of the probability of detection

The standard deviation σ of a Gaussian distribution is calculated for example by using Formula (BB.3):

$$\sigma = \sqrt{\frac{\sum_{i=2}^n (F(z_i) - F(z_{i-1})) \times \left(\frac{z_i + z_{i-1}}{2} - \mu \right)^2}{F(z_n) - F(z_1)}} \quad (\text{BB.3})$$

$F(z_i)$ is the ratio of the number of measurements with the test piece detected as being inside the detection zone to the number of all measurements at the position z_i .

μ , the maximum of the Gaussian distribution ($F(\mu) = 0,5$) is also a numerical value calculated for example by using Formula (BB.4):

$$\mu = \frac{\sum_{i=2}^n (F(z_i) - F(z_{i-1})) \times \left(\frac{z_i + z_{i-1}}{2} \right)}{F(z_n) - F(z_1)} \quad (\text{BB.4})$$

This measurement is performed with the test piece positioned in the image corner at the maximum operating distance.

NOTE 1 The use of numerical integration techniques such as Simpson's rule could be appropriate.

NOTE 2 For determining STZ refer to Clause BB.5.

BB.4 Determination of tolerance zone for systems providing distance information

Systems that provide a distance value to the objects in the detection zone allow a direct determination of the tolerance zones by analysis of the measurement accuracy.

The measurement accuracy can be determined by performing repetitive position measurements of a test piece and then using statistical analysis to derive the statistical and systematic measurement errors of the system. From these measurement errors the corresponding tolerance zones (PTZ and STZ) can be calculated.

The appropriate test setup for this measurement is the setup for low contrast (LC). A grey test piece (GTP) according to the specified detection capability of the sensor shall be positioned in front of a grey background (GB) with at least 50 cm between test piece and background. The distance between sensor and test piece shall be the maximum operating distance of the system. The test piece shall be positioned in the image corner and shall be oriented such that its axis is perpendicular to the z-axis of the sensing device coordinate system.

Repetitive distance measurements of the test piece in the specified test setup are performed. A set of 1 000 distance measurements can be analysed regarding mean value μ and standard deviation σ with sufficient statistical accuracy. It can now be inferred from Formula (BB.2) that the size of the PTZ is five times the standard deviation σ . Furthermore, the systematic measurement error can be determined: The deviation of the mean value μ from the real test piece position is regarded as a systematic measurement error and shall be added to the size of the STZ.

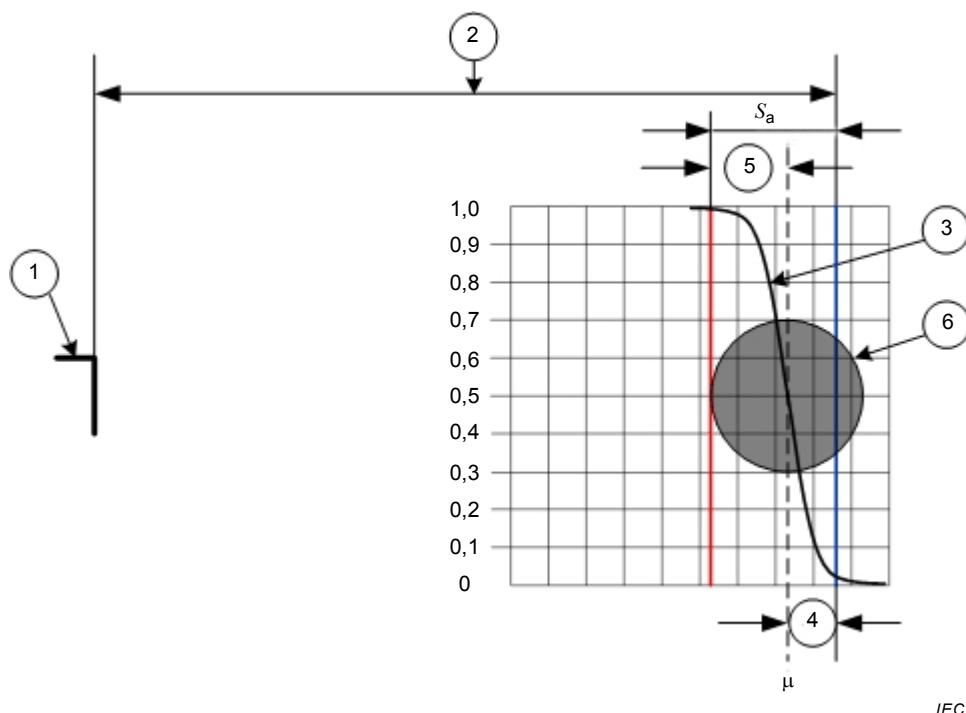
It is important to note that a common distance value attributed to the test piece as a whole shall be used for the determination of the tolerance zones. The standard deviations of pixel-based distance values that are averaged, filtered or in some other way processed by a

specific detection algorithm cannot be easily related to the detection probability and hence are not useful to calculate the sizes of tolerance zones.

BB.5 Tolerance zone related to systematic interferences

The tolerance zone S_a is also affected by influences that are not probabilistic. An additional systematic tolerance zone (STZ) is defined and added to PTZ (as shown in Figure BB.3).

STZ is the difference between the border of the detection zone and μ .



Key

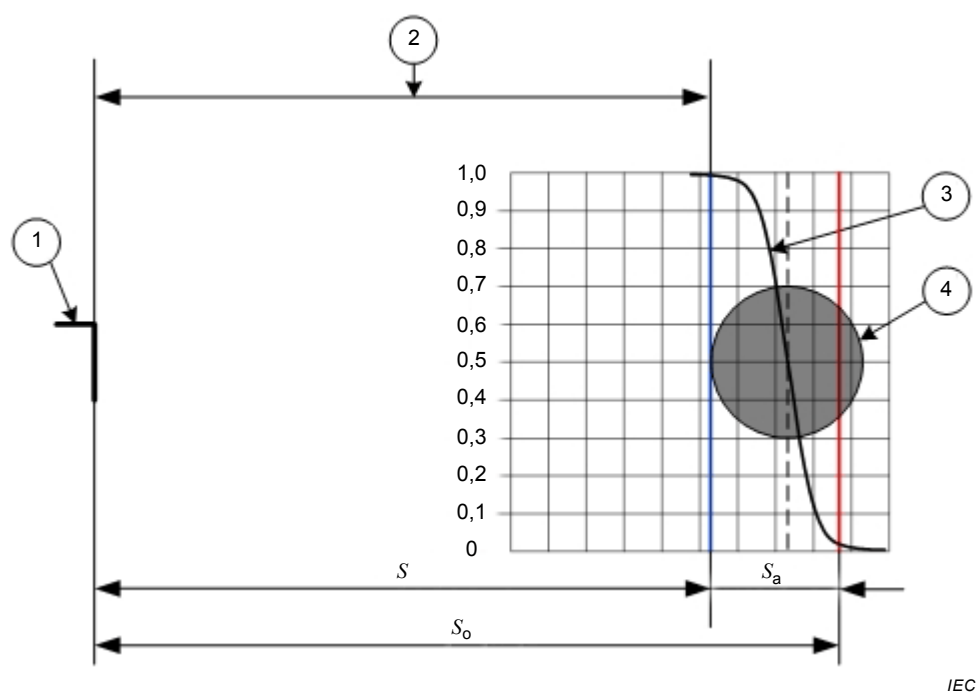
- 1 – Hazard zone
- 2 – Detection zone
- 3 – Probability of detection
- 4 – Tolerance zone related to systematic influences (STZ)
- 5 – Tolerance zone related to probability (PTZ)
- 6 – Test piece

Figure BB.3 – Relationship between detection zone and tolerance zone

NOTE Clause BB.5 and Figure BB.3 describe the procedure to determine the systematic part of the tolerance zone (STZ). Therefore, the tolerance zone is shown as part of the detection zone in Figure BB.3. Clause BB.6 describes how to use the measured values.

BB.6 Adding the tolerance zone on the outer border of the detection zone

The tolerance zone S_a has to be added to the outer border of the detection zone to ensure the required probability of detection inside of the detection zone (as shown in Figure BB.4).

**Key**

- 1 – Hazard zone
- 2 – Detection zone
- 3 – Probability of detection
- 4 – Test piece

Figure BB.4 – Overall minimum distance S_o including tolerance zone

Annex CC (informative)

Basic principles of physics for contrast of convex homogeneous bodies

CC.1 Illumination on a surface element

The brightness of a surface element, of any convex body with homogenous diffuse (Lambert's) reflectivity depends on the direction of illumination φ on the surface element (see Figure CC.1 and Figure CC.2) and is calculated according to Formula CC.1.

$$I(\varphi) = I \times \cos(\varphi) \quad (\text{CC.1})$$

In the case of a sphere illuminated by a point source (see Figure CC.1), the behaviour is as follows:

- The surface of the sphere normal to the direction of the point source ($\varphi = 0^\circ$) is illuminated with maximum intensity.
- The surface of the sphere parallel to the direction of the point source ($\varphi = 90^\circ$) is illuminated with minimum intensity (i.e. dark).
- The backside ($\varphi > 90^\circ$) of the sphere not in direct view of the point source is dark.

It is possible to use a half-Ulbricht sphere model to approximate the floor-level illumination of a large, high ceiling hall with lamps densely and equally distributed.

NOTE 1 This is not a worst case under some conditions. Additional analysis can be necessary by the supplier to identify critical situations based on specified lighting requirements and operating conditions.

In the case of a test sphere placed at the center of a half-Ulbricht sphere (see Figure CC.2), the brightness varies as follows:

- The top of the sphere is illuminated with maximum intensity.
- The side of the sphere ($\varphi = 90^\circ$) is illuminated with the half intensity.
- The back is dark.

NOTE 2 An Ulbricht sphere illumination doesn't appear in the real world. In case of any convex body illuminated from all directions (Ulbricht sphere), every surface element has the same brightness. No structure is visible, only different colours induce a contrast on the surface.

The brightness of a surface element of a spherical test piece is calculated by integrating all incidence angles of the surface element (see Figure CC.3) according to Formula CC.2:

$$I(\varphi) = I \times \frac{(1 + \cos(\varphi))}{2} \quad (\text{CC.2})$$

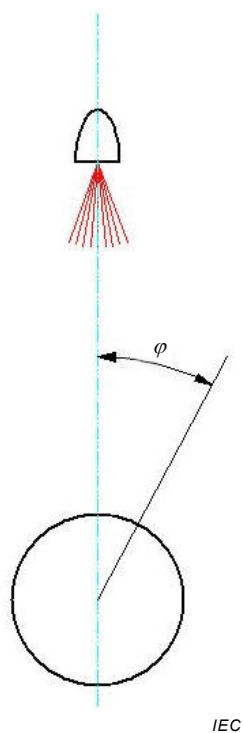


Figure CC.1 – Illumination model – Sphere illuminated by a point source

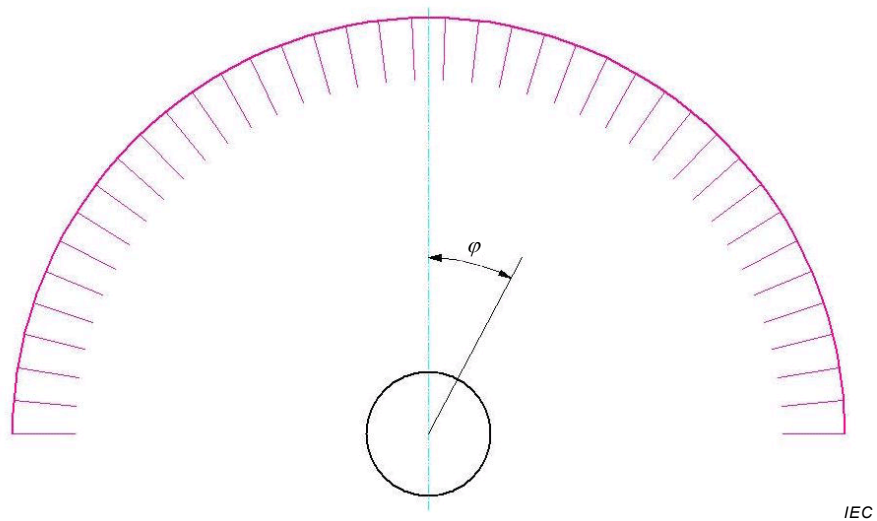


Figure CC.2 – Illumination model – Sphere illuminated by a half-Ulbricht sphere

CC.2 Brightness of a surface element

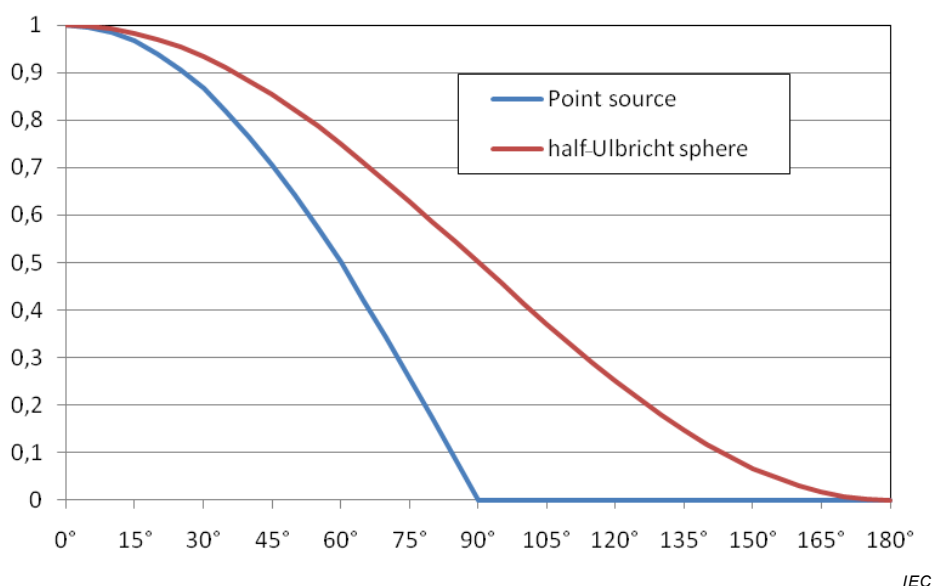


Figure CC.3 – Brightness of a surface element of a sphere in spherical coordinates

When a spherical test piece is illuminated by half of an Ulbricht sphere and projected on an image (parallel projection), the intensity profile in the image depends on the viewing direction (see Figure CC.4).

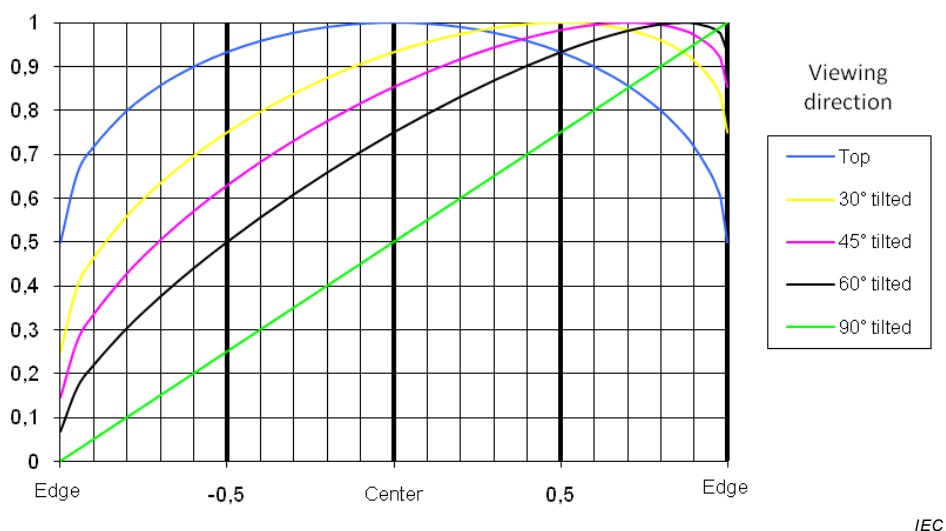


Figure CC.4 – Brightness distribution in an image of a sphere

The contrast is as low as possible when the sphere is viewed from the same direction as the illumination. The contrast in this specific lighting plan and test piece position reaches a theoretical minimum when the average grey values with or without the test piece are the same (see Figure CC.5).

NOTE A practical test setup can alternately use a test piece and background of the same reflectance in worst case lighting conditions, as established through analysis (see 5.1.2.5, Setup for low contrast (LC)).

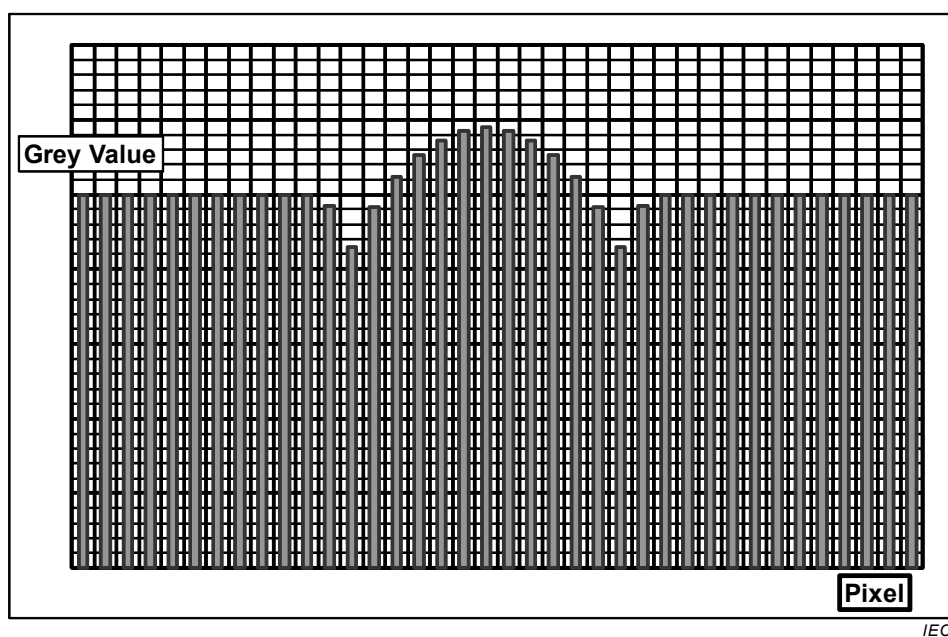


Figure CC.5 – Grey value profile over a sphere with low contrast for a typical imaging contrast (Modulation Transfer Function)

Grey value distributions shown in Figure CC.6 to Figure CC.8 have better contrast than the distribution shown in Figure CC.5.

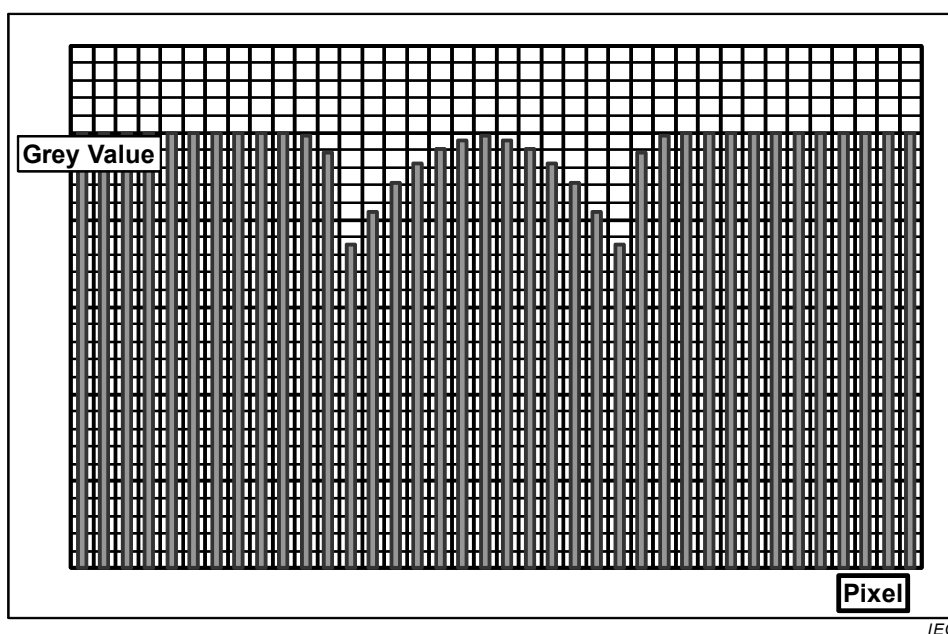


Figure CC.6 – Grey value profile over a sphere with the same colour as the background

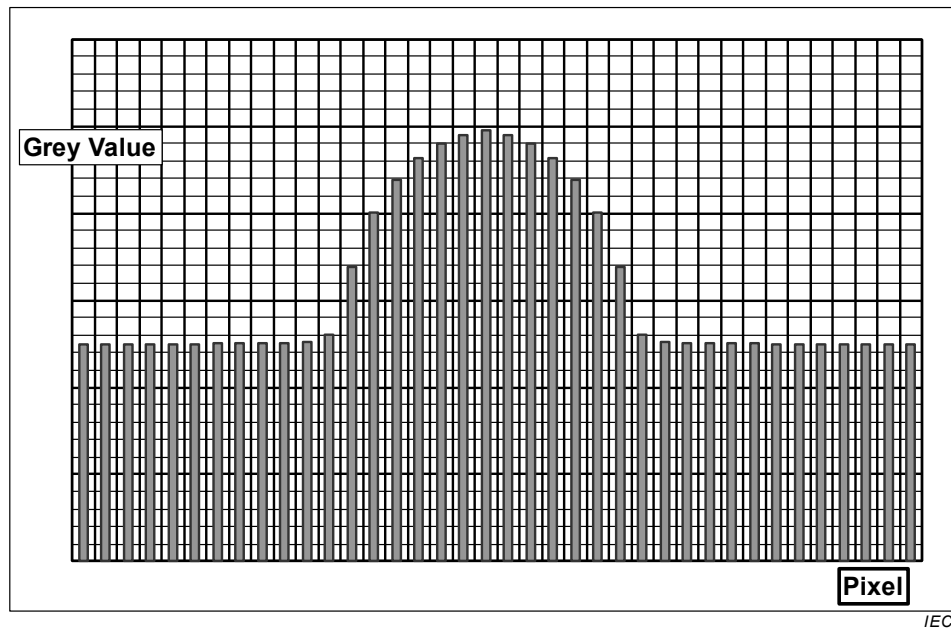


Figure CC.7 – Grey value profile over a sphere in front of a background that is half as bright

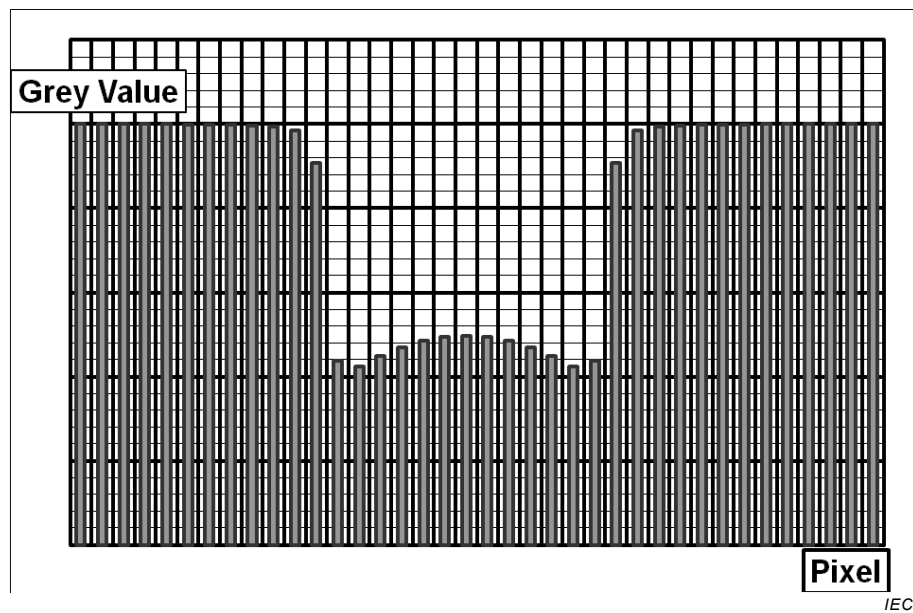
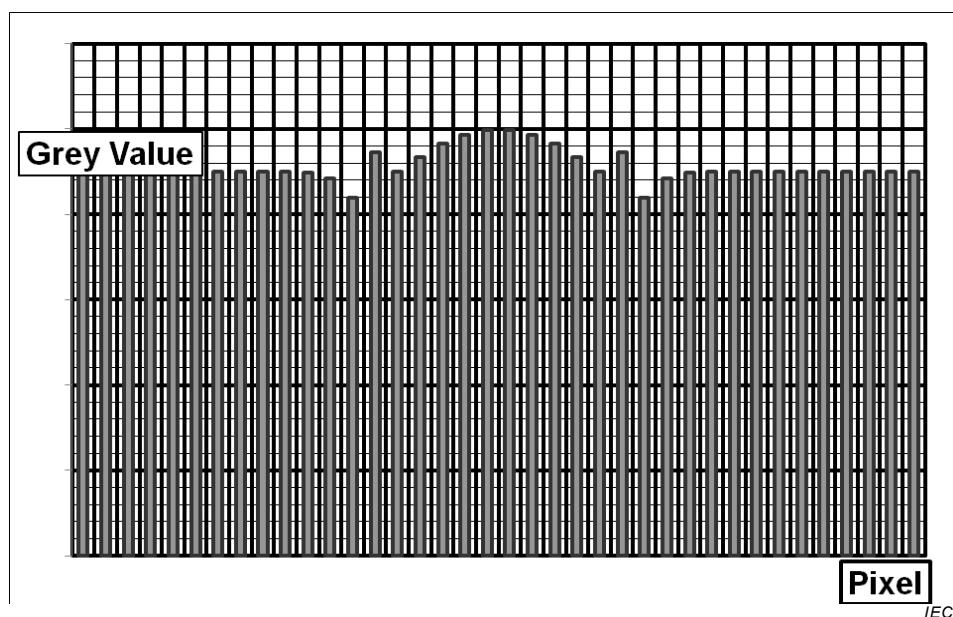


Figure CC.8 – Grey value profile over a sphere in front of a background that is twice as bright

Figure CC.9 and Figure CC.10 show two other worst case situations.



NOTE The side of the sphere is 1,2 times brighter than the top (like dark hair and white skin).

Figure CC.9 – Grey value profile over a sphere by low contrast

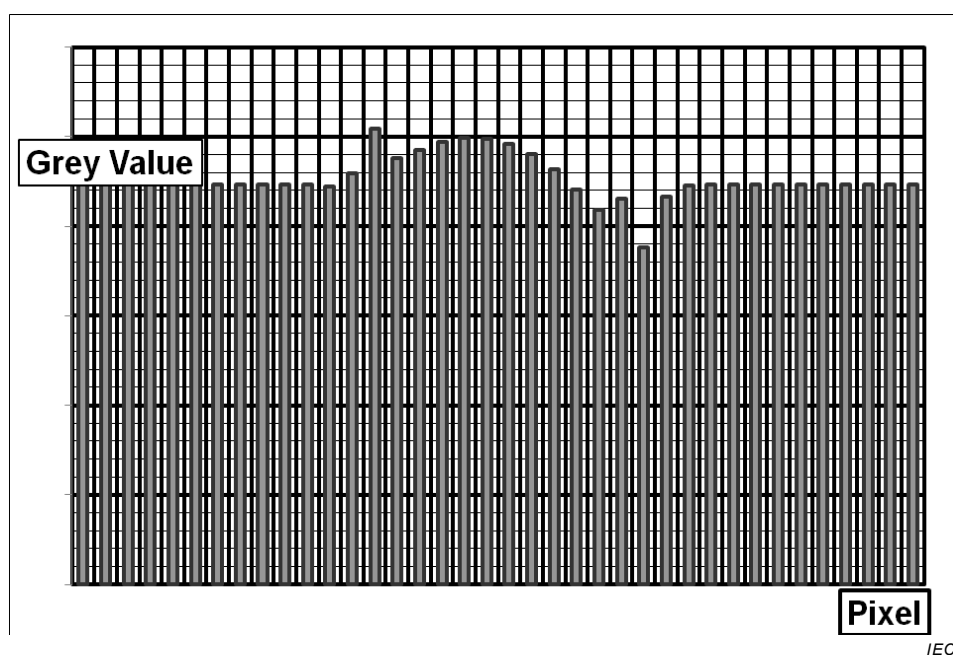
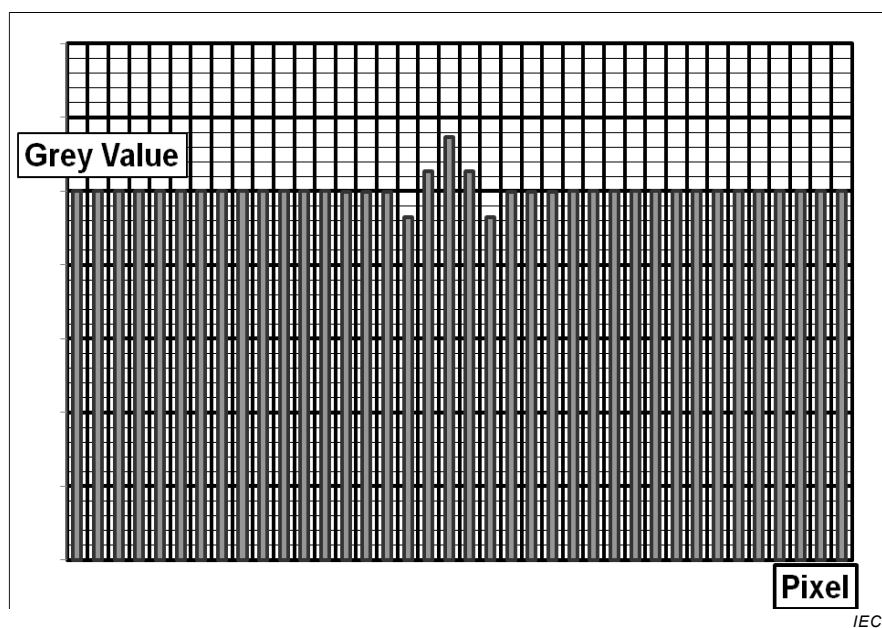


Figure CC.10 – Grey value profile over the sphere from Figure CC.9 but with the direction to the imaging device changed by 10°

Figure CC.11 shows a sphere with minimum size which leads to an object with a 5 pixel diameter in the image plane.



IEC

Figure CC.11 – Grey value profile over a small sphere that results in an image that is 5 pixels in diameter

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